Day 7 Data Report

Multi-sensor self-monitoring, 14 October 2026,
with comparison to days 1–6

Participant: Hemel

Analysis generated with `pipeline/daily_analysis.py`,
`compare_days.py` and `relationships.py`

1 Summary

Day 7, the last day of the study, included a short bedroom night, a morning trip to the lab (stairs, walk, rickshaw, lift), about 1.5 h in the lab with the environment node, posture cushion and, for the first time, a streaming smart bottle, an afternoon of sitting, and an evening walk with both insoles. The participant drank water before both walks and made a known 86 g test pour into the smart bottle. Table 1 lists the headline metrics; Sections 11 and 13 compare all seven days. The main findings are:

- **Smart bottle validated.** The test pour of 86 g at 11:52 was measured as +84.1 mL by the bottle's fused estimate (the main channel; −2.2 %). The capacitive level sensor read 70 mL (−18.5 %) and is usable for changes once its invalid 500 mL readings (16 % of samples) are removed. The bottle's own data stream detected two drinks, 121 mL at 11:34 and 34 mL at 11:56, which the participant confirmed; they are added to the bottle log, giving 1,265 mL for the day.
- **Drinking before the walks.** 280 mL at 10:10 and 500 mL at 19:00. The evening walk had the lowest walking heart rate of the study (median 92 bpm) and the only one below the resting value before it. Over all ten walks with a valid heart rate, however, the water drunk in the 2 h before a walk was not related to walking heart rate ($\rho = 0.27$, CI −0.59 to 0.83; Section 12).
- **Heart rate and step rate** remain positively related on all seven days (pooled $r = 0.29$). The relationship is reproducible but weak (8 % of the variance) and appears only over windows of 10 minutes or more (Section 13).
- **Right heel sensor fault** during the evening walk was detected automatically (dropout in 27 % of the moving minutes, mostly 19:07–19:18). Stance and swing from IMU gait events stayed valid: right stance 63.6 %, left 62.4 % (369 paired strides), push-off timing symmetric (+0.1 %), step timing 49.2 %.
- **Interesting findings** (Section 7): the *slip* at 19:21 is the largest right-foot acceleration of the walk (79.7 m/s^2 at 19:21:02, more than twice a normal heel strike), and *soldering-iron fumes* at 19:10 drove $PM_{2.5}$ above the sensor's ceiling of 3,333 $\mu g/m^3$ within 20 seconds, the highest particle reading of the study.

- **Sleep** was the shortest of the study: 5.4 h (04:35–10:03).
- **Bedroom** $PM_{2.5}$ again fell during the night (about 110 to 40 $\mu g/m^3$ between 03:00 and 05:30), the sixth monitored night with this shape, though at a lower level than before.

Table 1: Headline metrics of day 7 (14 October 2026).

Domain	Metric	Value	Unit
Sleep	Time asleep	5.4	h
	Sleep efficiency	98	%
	Light / deep / REM	47 / 30 / 23	%
	Mean HR asleep	64	bpm
	Mean SpO_2 asleep	97.2	%
Activity	Steps (band / phone)	4,284 / 4,694	
	Mean awake HR	96	bpm
	Minutes with HR $>$ 120 bpm	24	min
	HR–steps Spearman ρ	0.08	
	HR lag behind steps	13	min
Gait (walk)	Cadence	100	steps/min
	Stride time / CV	1.20 / 15.4	s / %
	Stance, IMU (R / L)	63.3 / 62.5	%
	Swing, IMU (R / L)	0.438 / 0.465	s
	Stance, pressure (R)	–	%
	Push-off timing (R / L)	66.3 / 66.8	%
	Distance / speed	747 / 1.18	m / m/s
	Stride length	1.27	m
	Insole − band bias	+6.3	steps/min
Posture	Monitored time	22	min
	Upright / slouched	18 / –	%
	Median time between changes	3.4	min
Environment	Indoor CO_2 (median)	628	ppm
	Indoor / outdoor $PM_{2.5}$	40 / 57	$\mu g/m^3$
	Indoor temp. / RH	26.9 / 49	°C / %
	Indoor / outdoor sound	56 / 75	dBA
	Bedroom CO_2 / $PM_{2.5}$	488 / 60	ppm / $\mu g/m^3$
	Bedroom temp. / sound	30.6 / 62	°C / dBA
Left/right	Left lag behind right	49	% of cycle
	Foot-tilt range L − R	+2.0	°
	Stance / swing SI (IMU)	-0.1 / +5.1	%
	Stance % (L / R, IMU)	62.4 / 63.6	%
Hydration	Total intake (drinks)	1265 / 6	mL
	Longest gap between drinks	7.6	h

2 Data collected

Table 2: Recording sessions on day 7 (local time, UTC+6), split by activity using the participant's notes, GPS speed and insole motion.

Key	Session	Time	Sensors	Notes
O	Bedroom	01:33–10:04	Environment node	Whole sleep period
S	Stairs and walk	10:54–10:58	Insoles, environment, GPS	Stairs at 10:54
K	Rickshaw ride	10:58–11:03	Environment, GPS	722 m at 4.2 m/s
A	Walk to the lab	11:03–11:09	Insoles, environment, GPS	Lift 11:05–11:06
L	Lab	11:09–12:46	Insoles (to 12:06), environment, smart bottle, cushion	Bottle test pour 11:52
P	Afternoon sitting	14:35–19:03	Posture cushion	Cushion connected 14 min
N	Evening walk	19:08–19:42	Insoles, environment, GPS	Slip 19:21; right heel fault

The smart bottle's log has drinks at 01:00 (100 mL), 02:32 (230 mL), 10:10 (280 mL) and 19:00 (500 mL), and its data stream recorded two more at 11:34 (121 mL) and 11:56 (34 mL), 1,265 mL in total. (As on every day of the study, the drink times and volumes come from the bottle's own log, read off the device.) The 86 g at 11:52 was a test pour into the bottle, not a drink. The left insole was disconnected from 19:18:52 to 19:22:26 during the evening walk; strides spanning the gap were removed and the rest of the walk was analysed. The wrist-band export (458 duplicated rows removed) runs until 21:11. Temperatures include the −5 °C correction. The right first-metatarsal sensor is still the replacement type, so stance and swing come from IMU gait events (see the day-6 report).

3 Smart bottle

The bottle node streamed to the app for the first time, from 11:14 to 12:46 (Fig. 1). It reports three channels: a load cell that weighs the content, a capacitive level sensor, and a *fused* estimate that the bottle uses as its main reading (the load cell while the bottle stands on it, a capacitive estimate while it is lifted). To check them, the participant poured a known 86 g of

water into the empty bottle at 11:52, and later confirmed the two removals described below as drinks.

Method. The pipeline looks for stable stretches of the fused signal (at least 20 samples within 3 mL) and takes the change between consecutive stable levels as a content change. Increases are fills or pours and are never counted as intake. Decreases are drinks. Stretches recorded while the bottle was off the load cell are not resting levels and are skipped, and each drink is timed at the moment the bottle was lifted. Capacitive readings of exactly 500 mL are the sensor's invalid value and are removed.

Result.

- *Fused estimate (main channel):* the test pour was measured as +84.1 mL (−1.9 mL, −2.2 %); the first reading after the pour was 85.5 mL. With the content unchanged for 49 minutes (11:57–12:46) it drifted by 5.1 mL (5–95th percentile), so drinks of about 10 mL or more are measured to within a few millilitres. The load cell alone gave the same values.

- *Capacitive level sensor:* once its invalid 500 mL readings (16 % of samples, 11:23–11:35) are removed, it read the pour as 70 mL (−18.5 %). It follows changes but reads about a fifth low, and it crept from 70 to 115 mL while the content stayed constant, so it is useful as a backup rather than for volumes.

- *Drinks detected by the bottle:* 121 mL at 11:34 (bottle lifted and emptied; during the lift the fused value fell 121 →73 →34 mL, showing the drinking) and 34 mL at 11:56 (lift, sip and put back). Both were confirmed by the participant and are added to the day's intake (open diamonds in Fig. 5).

The bottle streamed only during the lab period; outside it the drinks come from the bottle's own log. With the fused channel validated, keeping the bottle connected all day would give a complete, time-stamped intake record.

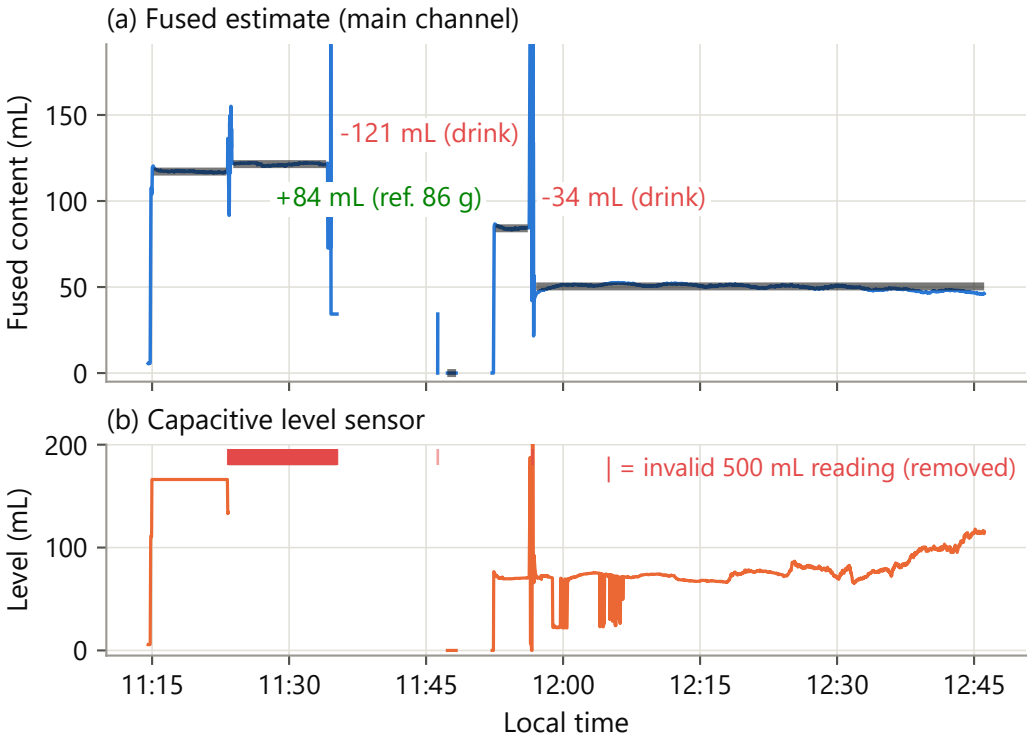


Figure 1: Smart-bottle node (11:14–12:46, 1 s samples; line breaks = disconnected). (a) Fused content estimate; dark bars: stable plateaus; labels: content change between plateaus (green: added, red: removed). Short spikes while the bottle was handled are clipped. (b) Capacitive level sensor; readings of 500 mL are the sensor's invalid value and were removed (16 % of samples). Test: 86 g poured at 11:52 was measured as 84.1 mL by the fused estimate (-2.2 %), 84.1 g by the load cell and 70 mL by the level sensor (-18 %). Drift at constant content (11:57–12:46, 5–95th percentile): fused 5.1 mL, level sensor 39 mL.

4 Heart rate, hydration and activity

Awake heart rate averaged 96 bpm with 24 minutes above 120 bpm (Fig. 2). The band counted 4,284 steps, the fewest of the study together with day 5. Heart rate rose during the morning trip (101–122 bpm, highest just after the lift) and was 100–112 bpm (median 106 bpm) while sitting from 13:00 to 19:00, a period with no drink between 11:56 and 19:00 (7.1 h). After the 500 mL drink at 19:00 it was 93–100 bpm before the walk and 90–96 bpm while walking at 64–108 steps/min. During a stop at 19:28–19:35 it fell to 76–88 bpm and it returned to 95 bpm when walking resumed, so the band

was following the pulse.



Figure 2: Wrist band over day 7: (a) heart rate (1 sample/min) and (b) steps per 15 min; sleep 04:35–10:03. Grey bands mark the sensor sessions: O overnight bedroom monitoring (01:33–10:04), S stairs and walk to the rickshaw (10:54–10:58), K rickshaw ride (10:58–11:03), A walk to the lab (with lift) (11:03–11:09), L lab (11:09–14:32), P afternoon sitting (posture only) (14:35–19:03), N evening walk (19:07–19:41).

Comparison with day 6. On day 6 nothing was drunk between 10:51 and 19:30 and the evening heart rate stayed at 130–145 bpm. On day 7 the participant drank 500 mL just before the evening walk, and walking heart rate was the lowest of the study. Drinking about half a litre of water is known to lower heart rate slightly within half an hour in healthy young adults. However, the two evenings also differed in route, stairs, crowding, metro travel and temperature (29 °C on day 7, 31–33 °C on day 6). Section 12 tests the question across all walks of the study.

Heart rate was hardly related to step rate over the whole day ($\rho = 0.08$, against 0.23–0.33 on days 1–6; Fig. 4). Most awake minutes were sitting with high heart rate and few steps, and the one walk had a low heart rate, which

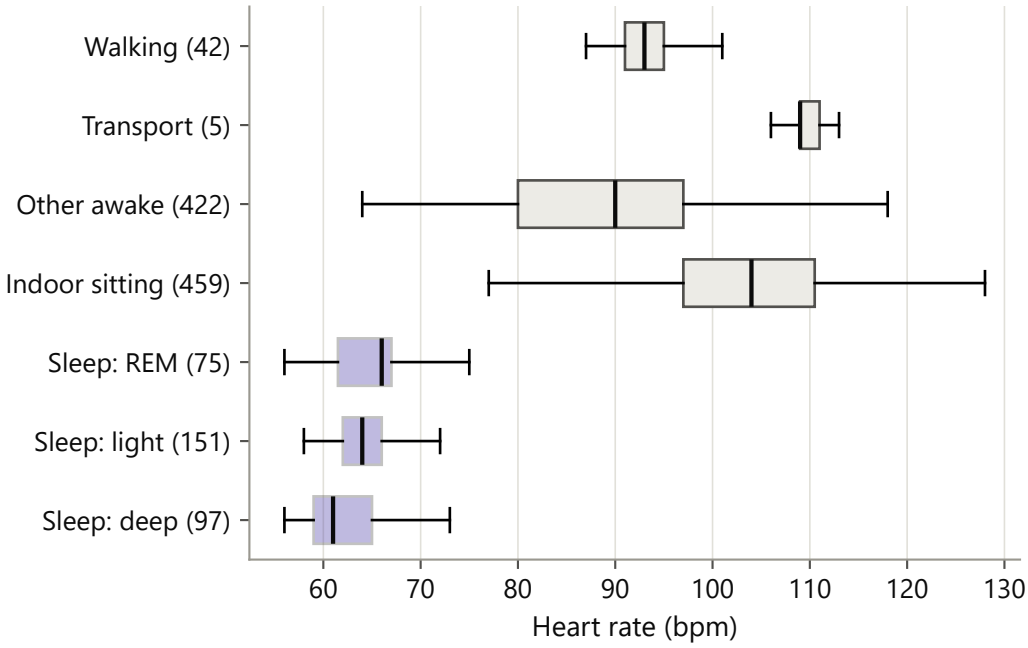


Figure 3: Heart rate by context over the day (minutes per context in brackets). Sleep stages come from the wrist band; awake contexts come from the sensor sessions (gait tests, indoor sitting with the environment node, walking sessions) and all remaining awake minutes. Boxes show the inter-quartile range and median; whiskers extend to 1.5 IQR.

together flatten the relationship.

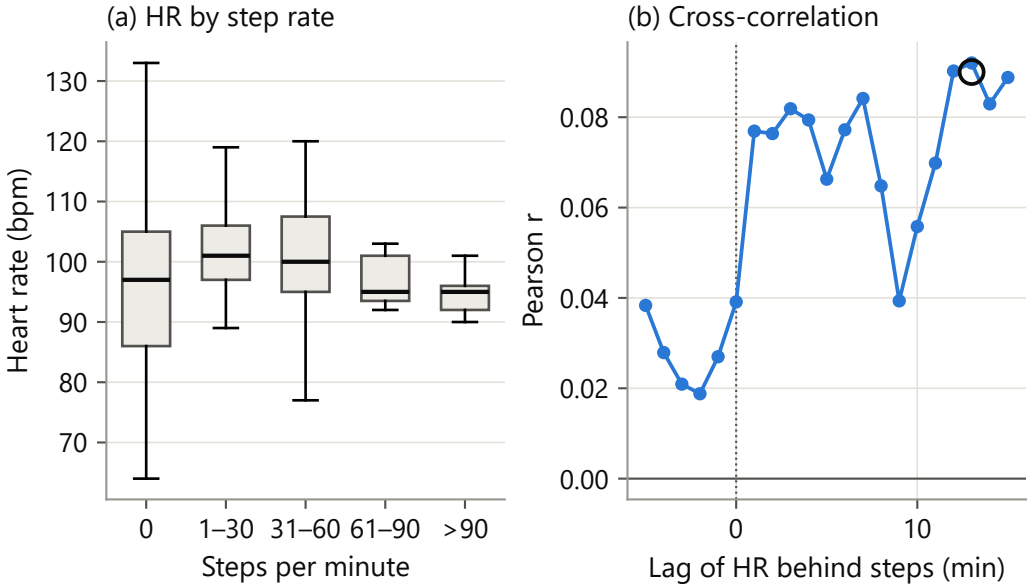


Figure 4: Heart-rate response to stepping (wrist band, 928 awake minutes). (a) Heart rate grouped by the number of steps in the same minute (minutes per group: 0: 829, 1–30: 46, 31–60: 28, 61–90: 11, >90: 14); Spearman $\rho = 0.08$. (b) Correlation between heart rate and step count shifted by -5 to $+15$ min; the strongest association ($r = 0.09$) occurs when heart rate lags steps by 13 min (circled). Minute samples are autocorrelated, so no p-values are reported.

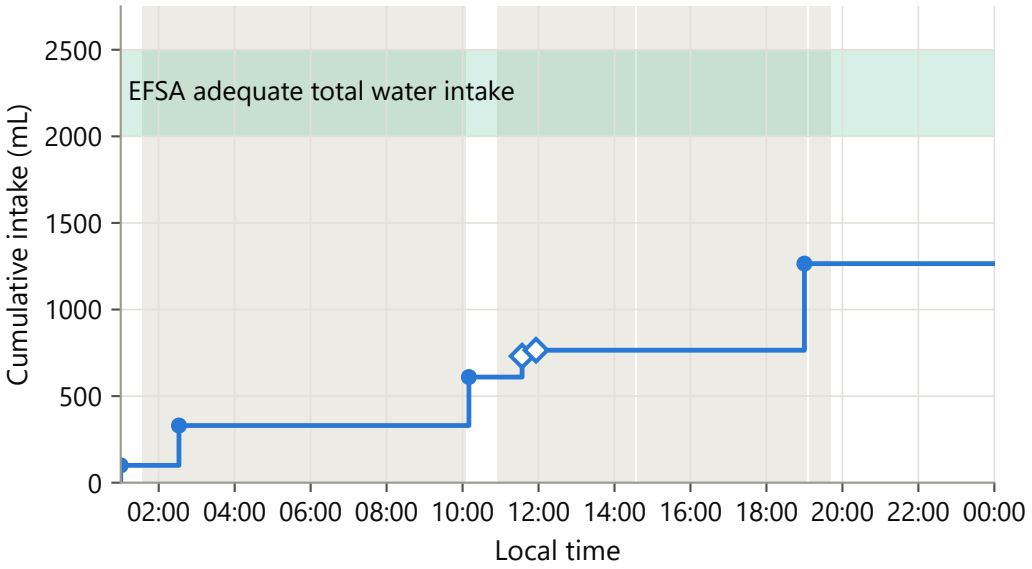


Figure 5: Cumulative fluid intake from the smart-bottle log (6 drinks, 1265 mL). Open diamonds: drinks detected in the bottle's data stream (155 mL). Grey bands mark the sensor sessions; the shaded band is the EFSA adequate intake for total water (2.0–2.5 L/day, including water from food). 610 mL was consumed before the first walk. Longest gap without drinking: 8 h (02:32–10:10).

5 Morning trip

The morning recording started at 10:54 with stairs and a short walk (S, 170 strides per foot, cadence 102 steps/min), followed by a rickshaw ride (K, 10:58–11:03, 722 m at 4.2 m/s) and a walk into the lab building with a lift ride at 11:05–11:06 (A, 73 left and 66 right strides). The rickshaw ride (Fig. 6) had $PM_{2.5}$ of 44–100 $\mu g/m^3$ and the loudest sound of the day (median 80 dBA). Strides inside the rickshaw and lift periods were excluded.

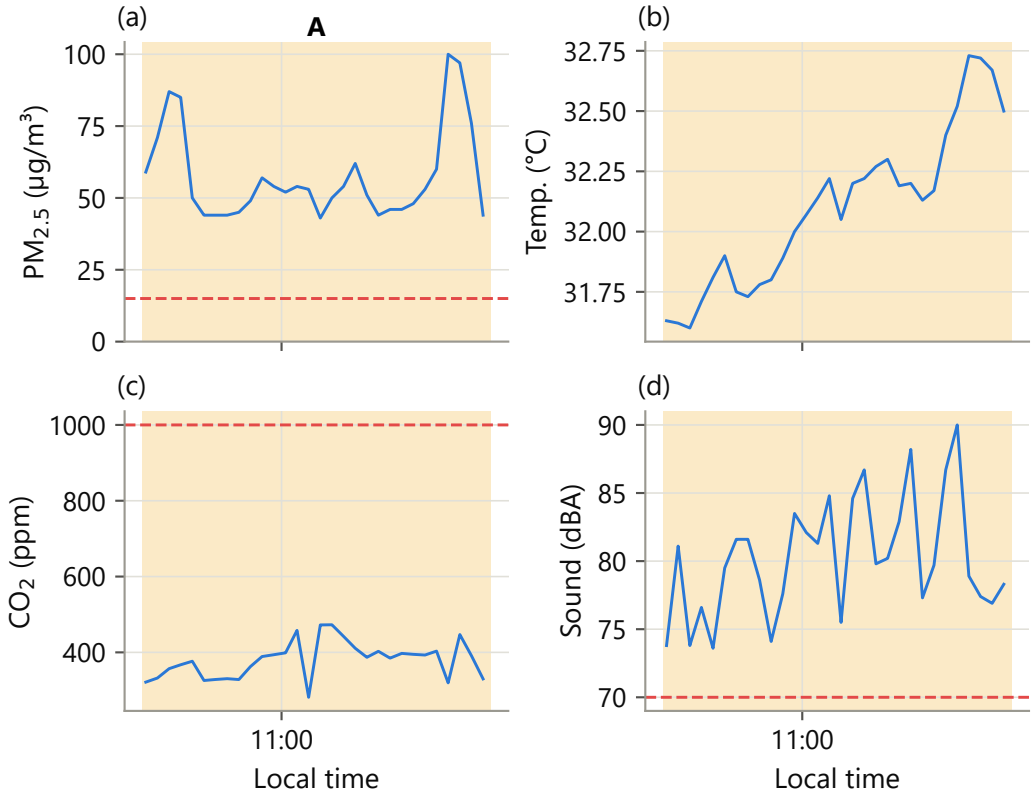


Figure 6: Environment node during session K (rickshaw ride, 10 s averages): (a) $PM_{2.5}$, median 52 $\mu g/m^3$, peak 100 (dashed: WHO 24-h guideline); (b) temperature, median 32.1 °C; (c) CO_2; (d) sound level, median 80 dBA (dashed: 70 dBA). Panel (c) shows CO_2 instead of humidity (median 388, maximum 473 ppm); the sensor reads low outdoors, so the rise inside vehicles is more reliable than the absolute level. Yellow bands mark noted events: A rickshaw ride (10:58–11:03). Temperature includes a -5 °C correction for sensor self-heating.

6 Evening walk and gait

The evening walk (N, 19:08–19:42) covered 747 m at a median 1.18 m/s (Fig. 7). It began indoors (temperature rising from 23 to 28 °C as the participant left an air-conditioned space), walking proper lasted from about 19:13 to 19:27 and 19:35 to 19:39, with a stop in between. The right insole detected 781 strides: median stride time 1.20 s, cadence 100 steps/min and stride length 1.27 m (Figs. 11 and 12). Stride-time variability (15 %) was higher than on most days because the walk included the indoor start and several stops.

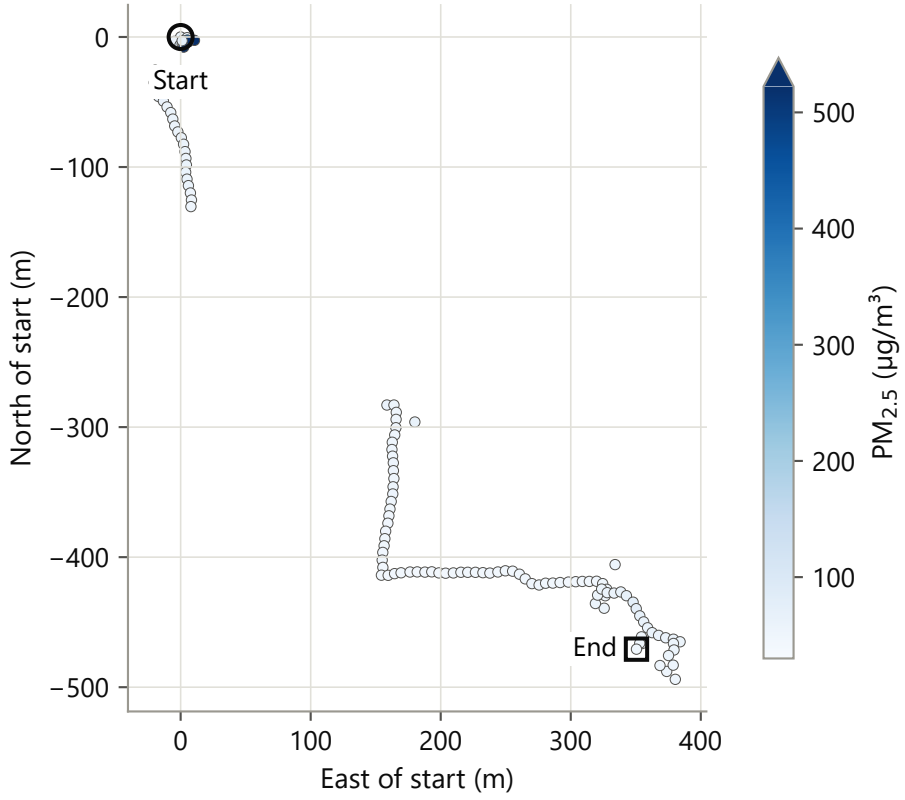


Figure 7: GPS track of session N (fixes with accuracy ≤20 m; logging began at 19:08): 747 m in 32 min at a median speed of 1.18 m/s, in metres from the start point, coloured by the $PM_{2.5}$ concentration measured by the body-worn environment node (colour scale capped at the 98th percentile, 522 $\mu g/m^3$; maximum 1659).

Environment. $PM_{2.5}$ had a median of 57 $\mu g/m^3$ and sound 75 dBA (Fig. 9), apart from the soldering-iron fumes at 19:10 described in Section 7. Strides during the slip at 19:21 were excluded from the gait results.

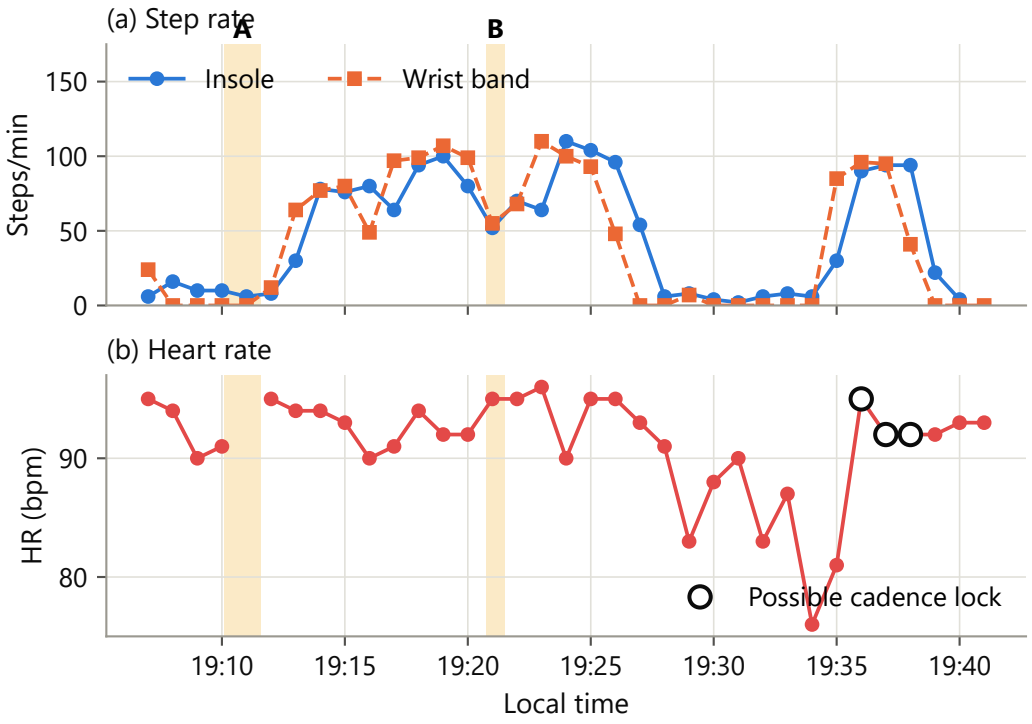


Figure 8: Session N (evening walk) minute by minute: (a) step rate from the insoles ($2\times$ the strides of whichever insole detected more in that minute) and the wrist band; (b) wrist-band heart rate (mean 91 bpm). Open circles: 3 minutes in which the band's heart rate was within 6 bpm of the step rate for at least two consecutive minutes, a sign that the optical sensor may have locked onto the walking rhythm instead of the pulse. Yellow bands mark noted events: A soldering-iron fumes (19:10–19:11); B slip (19:20–19:21).

Over the 99 minutes with insole data the insoles counted 2,866 steps and the band 2,244 (per-minute $r = 0.78$; Fig. 13).

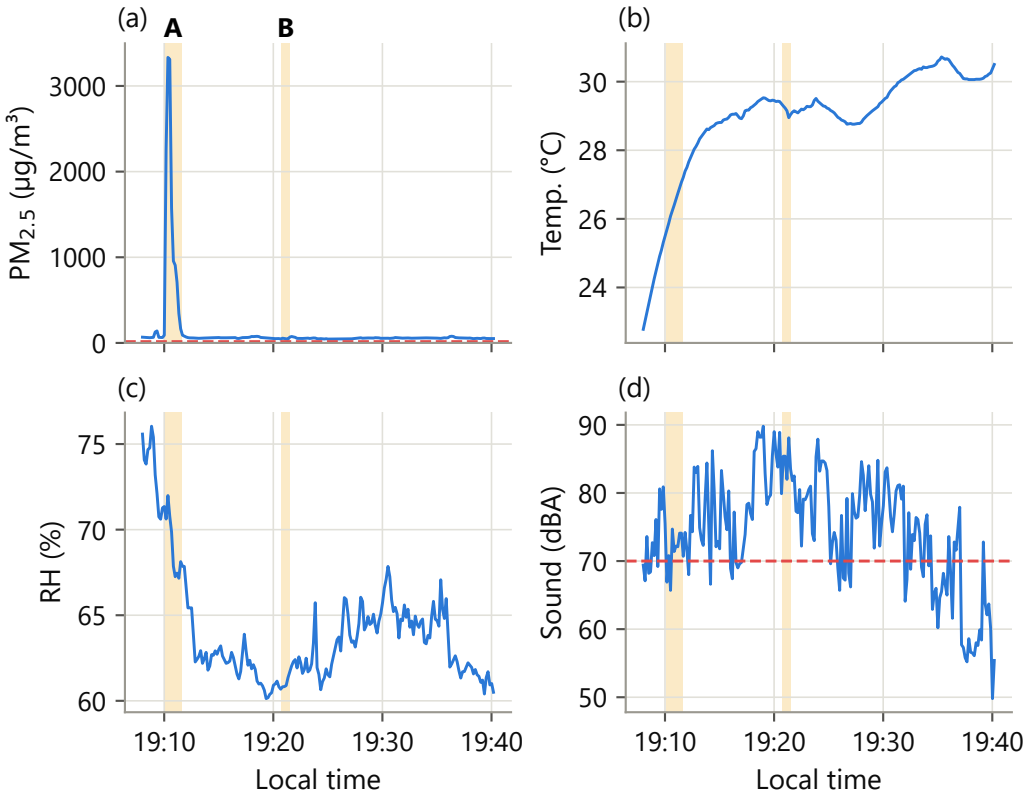


Figure 9: Environment node during session N (evening walk, 10 s averages): (a) $PM_{2.5}$, median 57 μg/m^3, peak 3333 (dashed: WHO 24-h guideline); (b) temperature, median 29.2 °C; (c) relative humidity, median 63 %; (d) sound level, median 75 dBA (dashed: 70 dBA). CO_2 is omitted because readings fell below the physically plausible outdoor level of about 420 ppm (minimum 364 ppm). Yellow bands mark noted events: A soldering-iron fumes (19:10–19:11); B slip (19:20–19:21). Temperature includes a -5 °C correction for sensor self-heating.

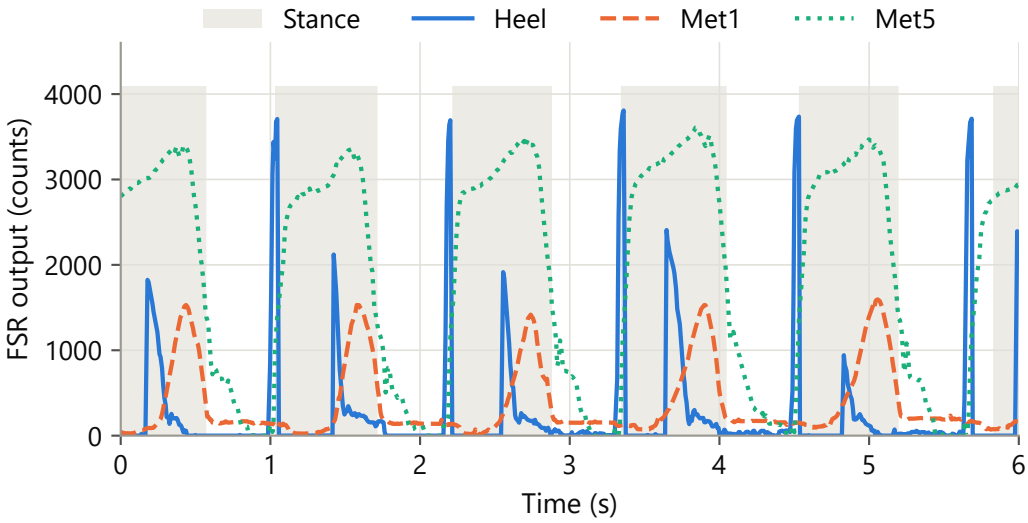


Figure 10: Right-insole plantar pressure during 6 s of steady walking (session N, evening walk). Lines show the heel, first-metatarsal (Met1) and fifth-metatarsal (Met5) force-sensitive resistors (12-bit ADC counts; higher means more load). Shaded intervals are stance phases detected by hysteresis thresholding of the summed, 6 Hz low-pass filtered load.

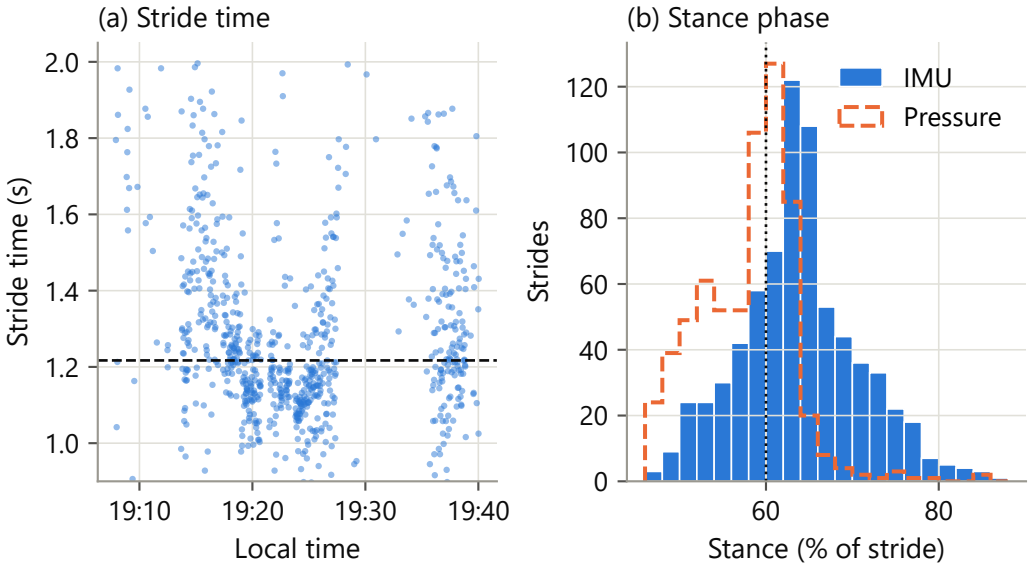


Figure 11: Right-foot stride parameters, session N (781 strides). (a) Stride time of every detected stride; dashed line = median (1.22 s, cadence 100 steps/min). (b) Stance phase as a percentage of stride time from IMU gait events (bars; heel-strike impact to IMU toe-off; median of steady-walking strides 63.3 %) and from the pressure sensors (dashed outline; 58.0 %) – not valid on this day because the first-metatarsal sensor had been replaced; dotted line = typical healthy value of about 60 %.

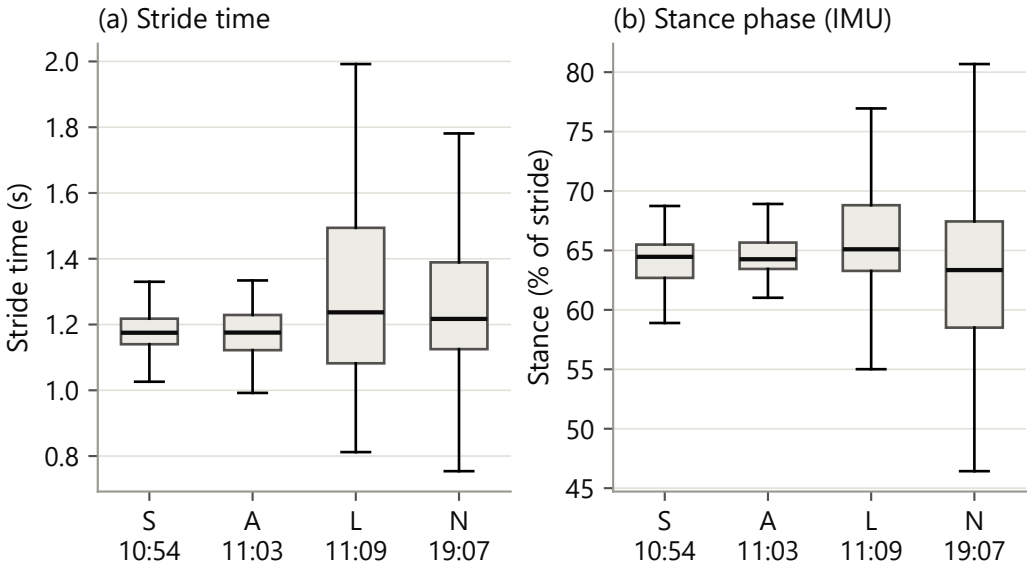


Figure 12: Right-foot (a) stride time and (b) stance phase (IMU gait events) across the gait recordings of day 7 (S: $n = 170$, A: $n = 66$, L: $n = 372$, N: $n = 781$; session start time below each key). Boxes show the inter-quartile range and median; whiskers extend to 1.5 IQR (outliers not drawn).

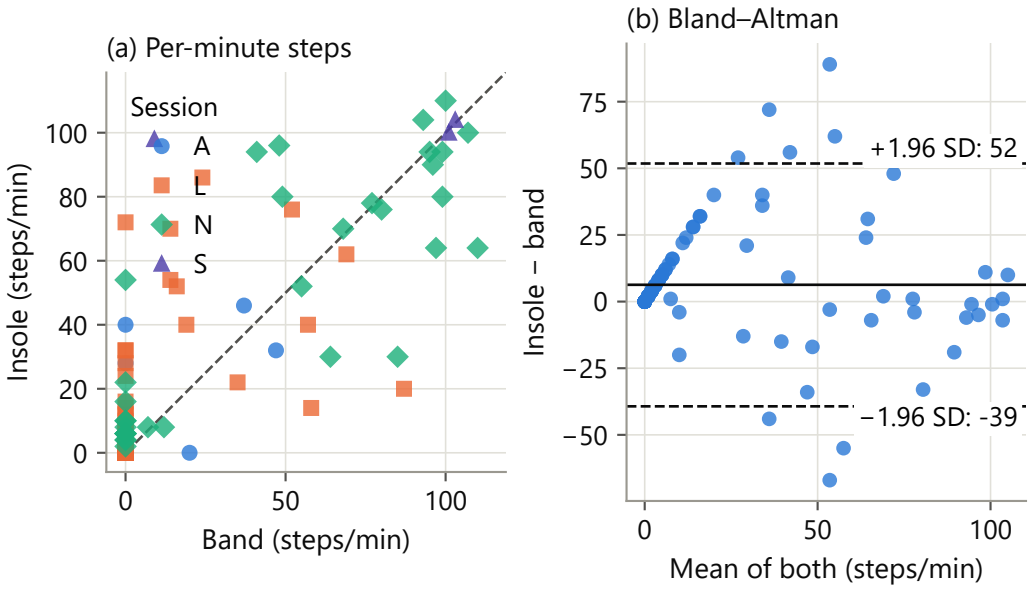


Figure 13: Agreement between insole-derived steps ($2\times$ the strides of the insole with more detections in each minute) and wrist-band steps for every minute of the gait sessions (99 minutes; totals 2866 vs 2244). (a) Per-minute counts with the line of identity ($r = 0.78$). (b) Bland–Altman plot: mean difference +6.3 steps/min (solid), 95 % limits of agreement -39 to 52 steps/min (dashed).

7 Interesting findings

A slip captured by the insole

The participant slipped at about 19:21 during the evening walk. The right insole recorded it clearly (Fig. 14); the left insole happened to be disconnected from 19:18:52 to 19:22:26.

- *Impact.* At 19:21:02 the right foot's acceleration reached 79.7 $\mathrm{m/s^2}$, the largest value of the whole walk. A normal heel strike in this walk peaked at 35.5 $\mathrm{m/s^2}$ (median), and 99 % of all 2-second windows stayed below 58 $\mathrm{m/s^2}$. The slip was therefore more than twice as sharp as a normal step.
- *Foot contact.* The step before the slip landed heel first (heel sensor about 2,600 counts). In the slip step both forefoot sensors loaded abruptly within 0.1 s (Met1 from about 100 to 1,160 and Met5 from about 700 to 3,320 counts) with no heel load, so the foot came down flat or forefoot first, as when a foot slides and is caught (panel c). The push-off rotation of that stride was smaller than usual (peak about 35° instead of 45–52°; panel b).
- *After the slip.* In the 15 seconds after it the right foot's tilt range was 44–50°, against 48–54° just before, which fits a more careful push-off. Over the following 2 minutes stride-time variability rose from 11.8 to 15.4 %, while median stride time (1.15 vs 1.17 s) and stance (63 % of the stride) were unchanged (panel d). Within a minute the walk was back to its normal pattern.

This shows that the insoles can detect a slip from a single acceleration peak far outside the normal range of the person's own steps, a possible basis for automatic slip or fall detection. With one event the detection threshold cannot be tuned; it would need more (safely simulated) slips.

Soldering-iron fumes

At 19:10, before the walk began, the participant was near a soldering iron in use. The environment node recorded the fumes at 10-second resolution (Fig. 15):

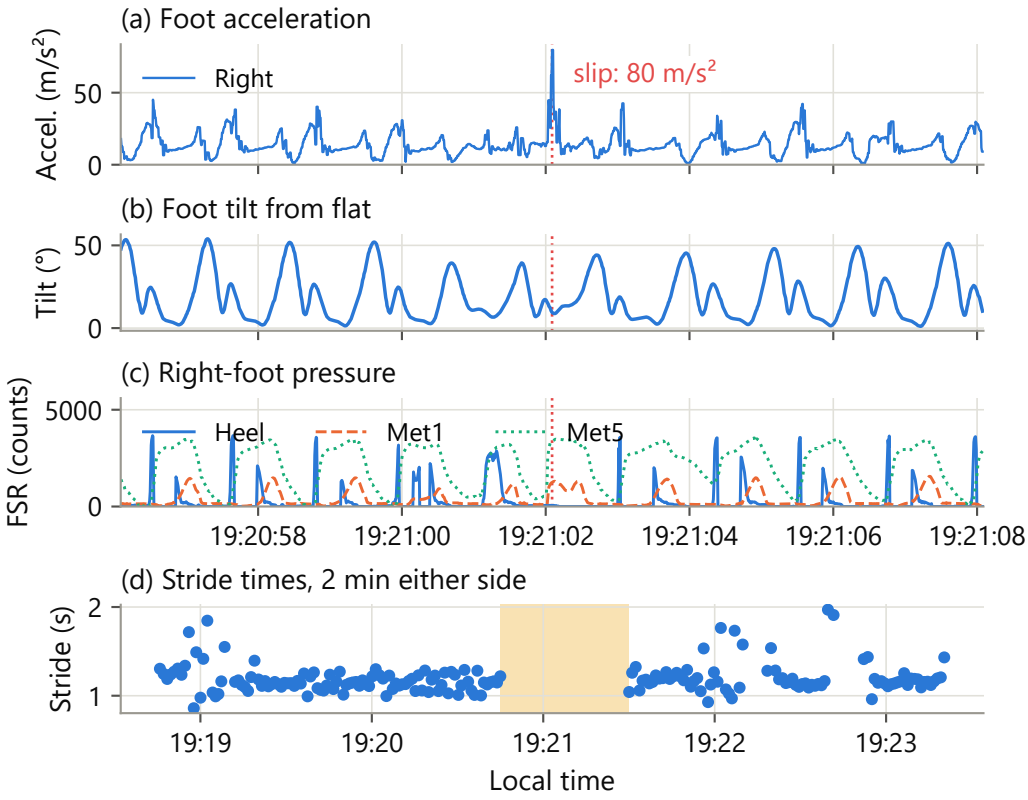


Figure 14: Close-up of the slip noted by the participant (session N). (a)–(c): 12 s centred on the largest acceleration inside the noted period (dotted line): (a) acceleration magnitude of each foot; (b) foot tilt away from the loaded flat position; (c) right-foot pressure sensors. (d) Stride time of every detected stride in the 2 minutes before and after; shaded: noted period (19:20:45–19:21:30), whose strides are excluded from the gait analysis. The right foot's acceleration peaked at 79.7 m/s^2 at 19:21:02, against a median heel-strike peak of 35.5 m/s^2 over the walk. A foot without a line was disconnected.

- *Size.* $PM_{2.5}$ rose from about 65 to 2,258 μg/m^3 within 10 seconds (19:10:10) and to 3,333 at 19:10:20. At that point PM_1, $PM_{2.5}$ and PM_{10} all read 3,333, the sensor's ceiling, so the true peak was higher. It is the highest particle reading of the study, about 4.7 times the day-6 smoke peak.

- *Particle type.* At the onset almost all of the mass was below 1 μm (PM_1 = $PM_{2.5}$ = PM_{10}). That fits soldering fumes, which are condensed rosin flux and are mostly sub-micron particles. As the plume aged, PM_1 fell first while PM_{10} stayed high for a further minute, consistent with

particles growing by coagulation.

- *Duration.* Concentrations were back to about 60 μg/m^3 by 19:12, so the exposure lasted about 90 seconds.
- *CO_2* rose from about 500 to 1,093 ppm at the same time. Heating flux releases very little CO_2, so this rise more likely reflects the participant breathing close to the node while working over the bench.

Brief, very high exposures like this one and the day-6 smoke event dominate short-term particle exposure but are invisible in daily or hourly averages. Soldering fumes are a recognised respiratory irritant, and local fume extraction at the bench is the standard precaution.

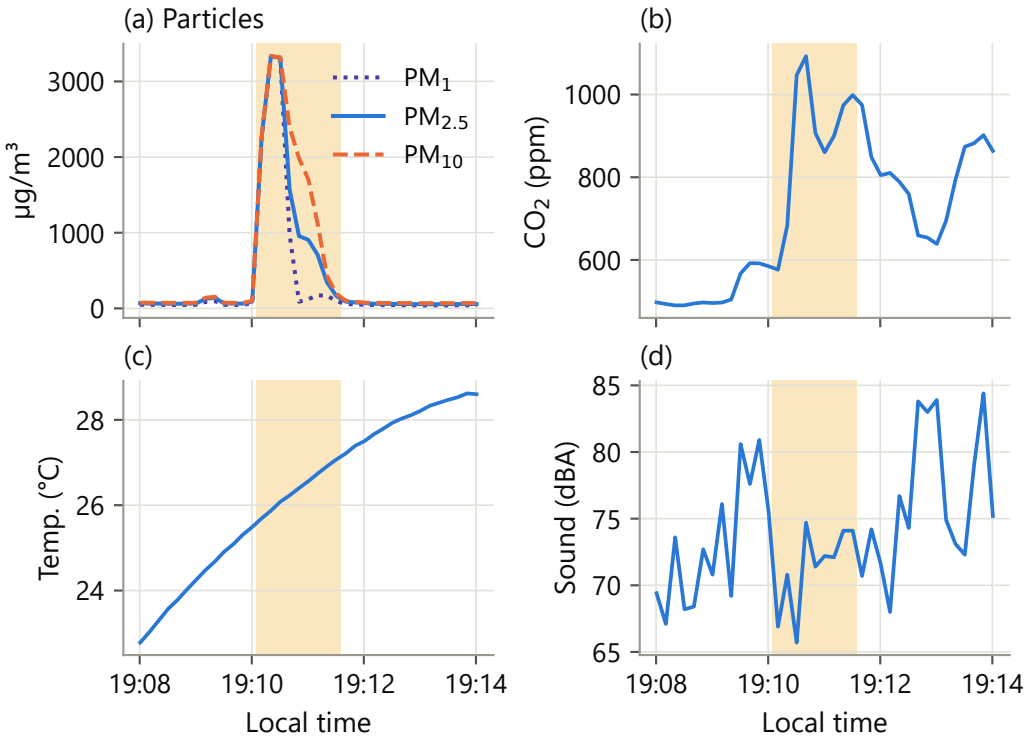


Figure 15: Close-up of the soldering-iron fumes (shaded, 19:10:05–19:11:35), environment node at 10 s resolution. $PM_{2.5}$ rose from about 65 to 3,333 μg/m^3 at 19:10:20 and was back at 58 μg/m^3 within minutes; CO_2 rose from 497 to 1,093 ppm. The three size fractions reached the same value (3,333), the sensor's ceiling, so the true peak was higher. Temperature includes a -5 °C correction for sensor self-heating.

8 Left versus right foot

Both insoles streamed at 112–114 samples/s (Fig. 16). The sensor-health check (Fig. 17) shows the right heel sensor dropping out in the first part of the evening walk (19:07–19:18, 27 % of the moving minutes), which matches the participant's report of a broken connection, and one dropout minute on the left heel. The pressure toe-off does not depend on the heel sensor, and stance and swing are taken from IMU gait events, so they remain valid.

In the evening walk 369 strides were paired (Fig. 18):

- stride time +1.1 % (1.222 vs 1.209 s), stance time −0.1 % (0.767 vs 0.768 s), stance 62.4 vs 63.6 % of the stride and push-off timing +0.1 %, all symmetric;
- swing time +5.1 % (0.461 vs 0.438 s), just above the ±5 % band;
- step timing from heel-strike impacts 49.2 % (Fig. 19) and 48.8 % from the cycle-averaged method (Fig. 20), i.e. symmetric;
- the foot-tilt range was *larger* on the left (+7.8 %), the reverse of days 1, 3, 5 and 6. The foot-tilt angle is measured from the foot's loaded, flat position, which is identified with the pressure sensors; with the right heel sensor failing, that reference is less reliable, so this reversal is probably a sensor effect.

The morning stairs-and-walk (S, 155 pairs) and the lab bouts (L, 150 pairs) gave stance 61.8 vs 64.6 % and 64.5 vs 64.1 %; S included stairs, which are not comparable with level walking.

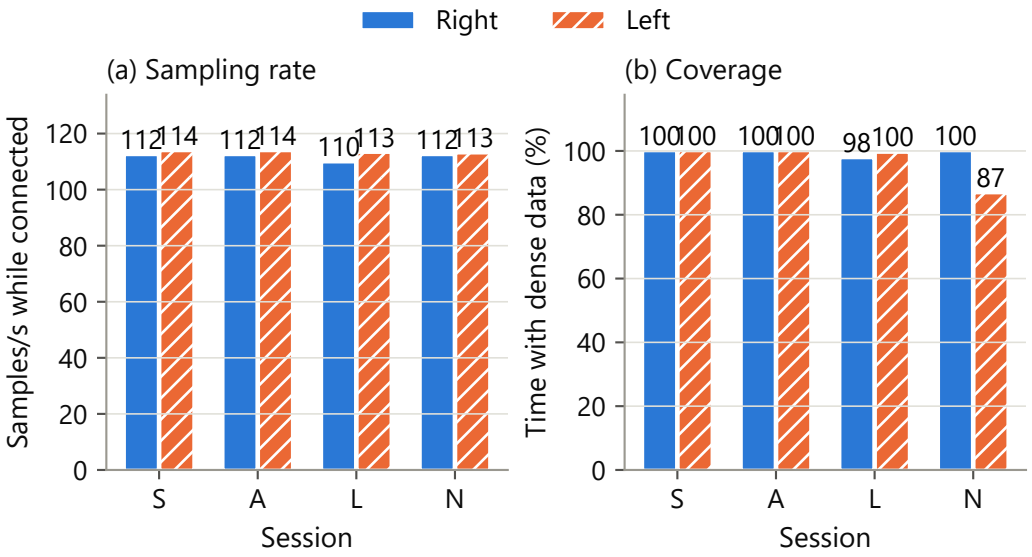


Figure 16: Data delivered by the right and left insoles in each session with both feet: (a) average sampling rate while the node was connected (nominal about 112 Hz) and (b) share of the recording covered by densely sampled data (consecutive samples $\leq$20 ms apart). The left insole's low coverage limits the left/right comparison to cycle-averaged timing and shape.

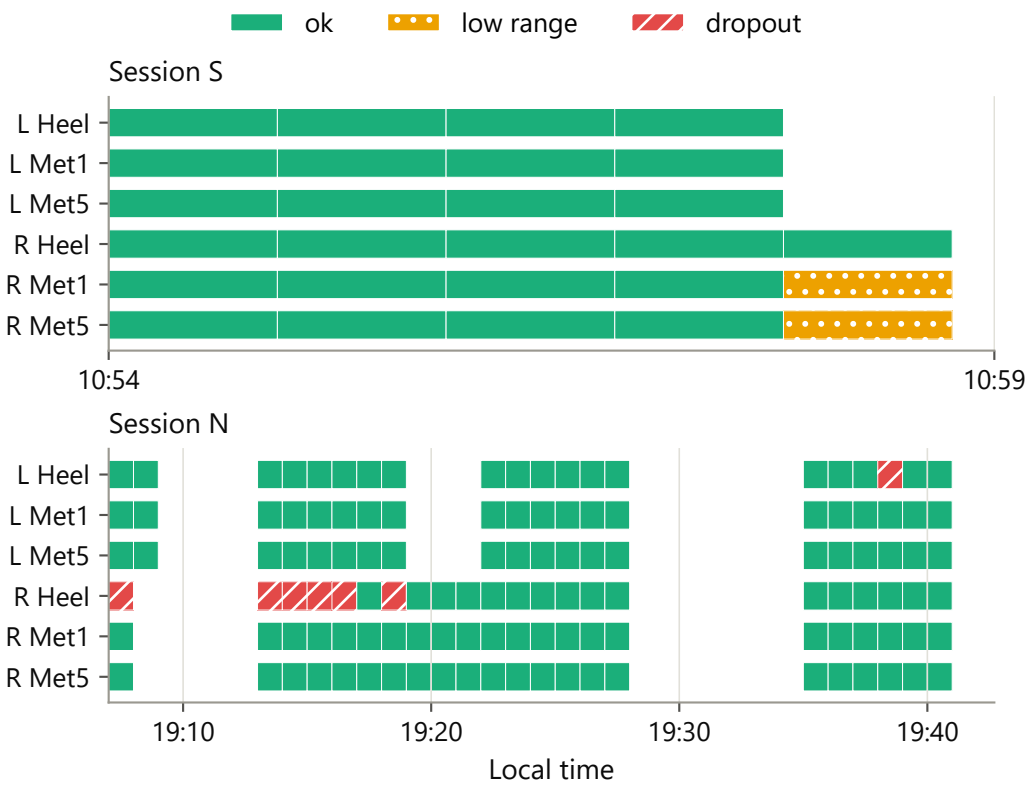


Figure 17: Status of each insole pressure sensor for every minute in which that foot was moving: OK, low range (signal present but its 95th percentile below 10 % of the foot's strongest sensor, e.g. a replacement sensor with a smaller output range) or dropout (no signal, or a full-range signal that is exactly zero in more than half of the samples, typical of a loose connection). Sensors faulty in at least 10 % of moving minutes: session S: R Met1 20 % faulty, R Met5 20 % faulty; session N: R Heel 27 % faulty.

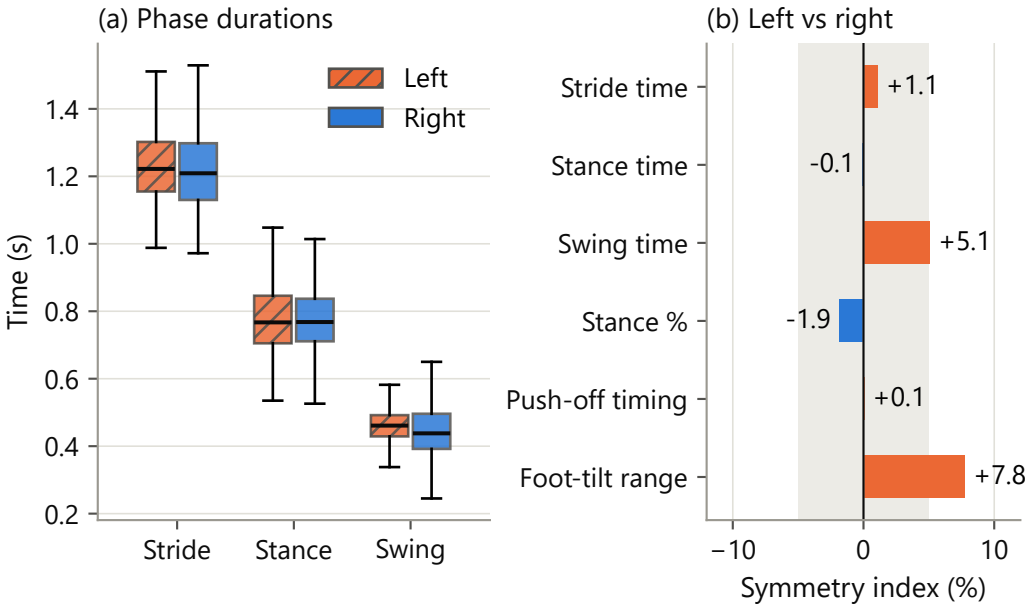


Figure 18: Stride-level comparison of the left and right foot during session N (369 paired strides: each right stride with the left stride that begins inside it; strides deviating more than 15 % from the local median were excluded as detection errors). (a) Stride, stance and swing durations per foot. (b) Symmetry index SI $= 100\,(L - R)/\overline{LR}$ of the medians (positive: larger on the left); the grey band marks ±5 %. Stance and swing are bounded by IMU gait events (heel-strike impact to IMU toe-off), defined identically on both feet and independent of the pressure sensors. Medians, left vs right: stride 1.222 vs 1.209 s, stance 0.767 vs 0.768 s (62.4 vs 63.6 % of stride), swing 0.461 vs 0.438 s, foot-tilt range 53° vs 49°. Pressure-based stance for comparison: 64.4 vs 58.1 % (right not valid: first-metatarsal sensor replaced by a different type).

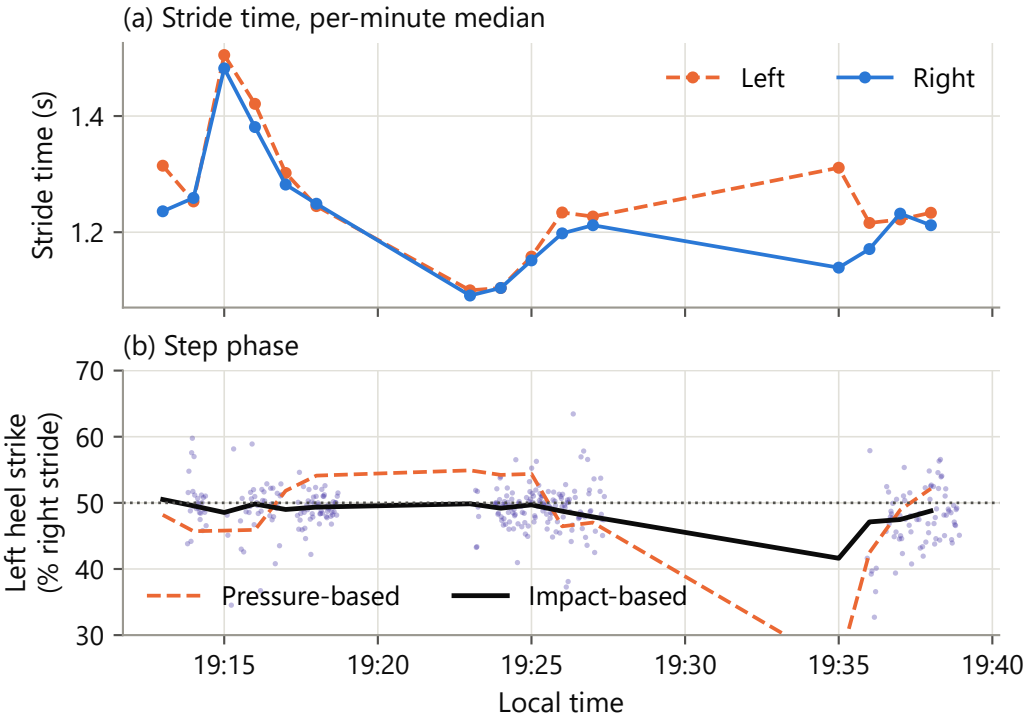


Figure 19: Left and right foot over session N. (a) Per-minute median stride time of each foot; the curves coincide, confirming that the two independently timed insoles agree (paired-stride correlation $r = 0.75$). (b) Timing of each left heel strike within the concurrent right stride with heel strikes taken from the accelerometer impact (dots and solid line: per-minute median; overall 49 %) or from the pressure sensors (dashed; overall 51 %); 50 % is symmetric. The pressure-based estimate drifts because the delay between pressure-detected heel strike and impact changed during the session on the left insole (45, 35, 31, 44, 45 ms in successive 5-min blocks; right: 11, -23, -16, 43, -14 ms). Clock drift is corrected; a constant radio-latency difference of a few tens of milliseconds between the nodes cannot be excluded.

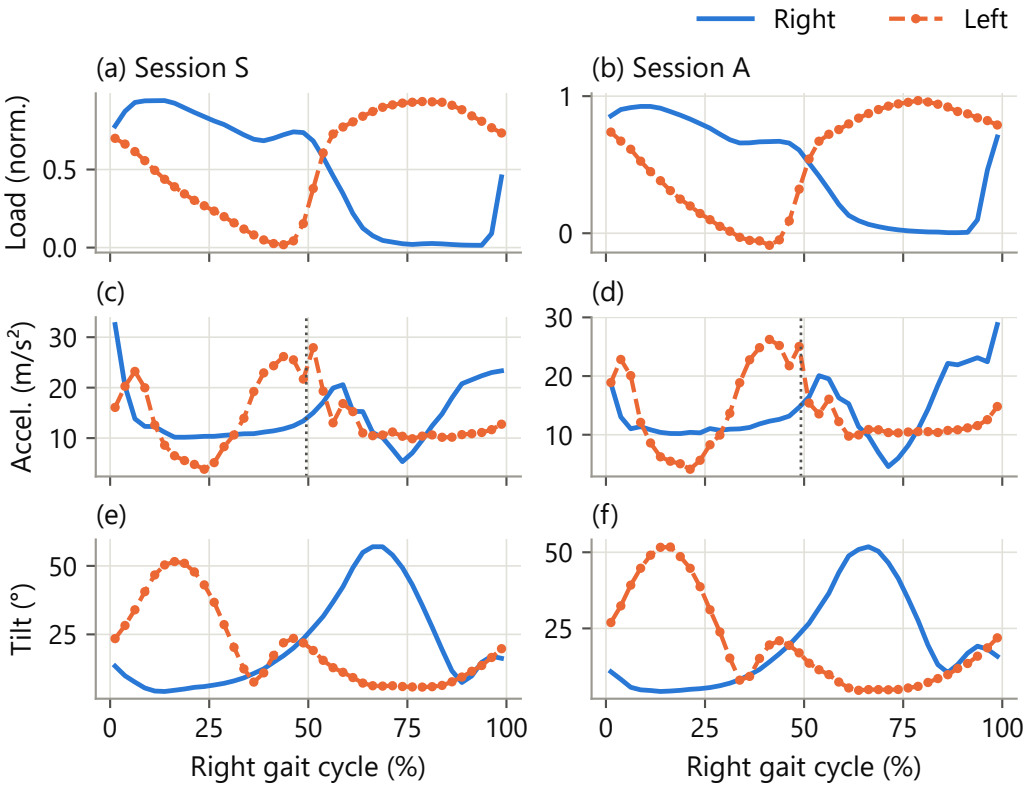


Figure 20: Left versus right foot over the right-foot gait cycle (0 % = right heel strike). Every left sample was placed at its phase within the concurrent right stride (steady strides of 0.8–1.6 s) and averaged in 40 phase bins (20 when data were sparse), so the method also works when the left insole delivers only bursts of data. Top: summed plantar load, normalised per insole. Middle: acceleration magnitude, whose heel-strike peak gives the left/right lag (dotted line; 50 % is symmetric) independently of how each insole's pressure sensors respond. Bottom: foot tilt from each insole's orientation, independent of sensor mounting. Results: session S: left impact lags right by 50 % of the cycle (95 % CI 49–50 %); load-based lag 55 %; tilt range L 46° vs R 53°; 22630 left samples in 169 strides; session A: left impact lags right by 49 % of the cycle (95 % CI 49–50 %); load-based lag 56 %; tilt range L 47° vs R 48°; 9689 left samples in 71 strides.

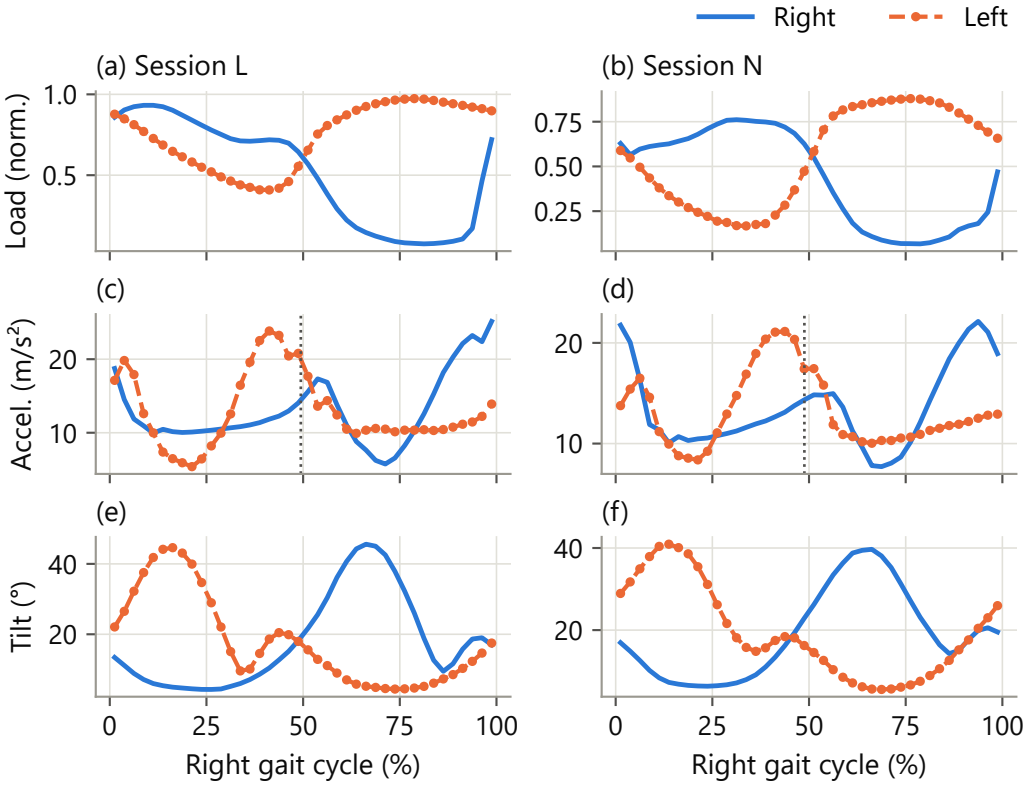


Figure 21: Left versus right foot over the right-foot gait cycle (0 % = right heel strike). Every left sample was placed at its phase within the concurrent right stride (steady strides of 0.8–1.6 s) and averaged in 40 phase bins (20 when data were sparse), so the method also works when the left insole delivers only bursts of data. Top: summed plantar load, normalised per insole. Middle: acceleration magnitude, whose heel-strike peak gives the left/right lag (dotted line; 50 % is symmetric) independently of how each insole's pressure sensors respond. Bottom: foot tilt from each insole's orientation, independent of sensor mounting. Results: session L: left impact lags right by 49 % of the cycle (95 % CI 49–50 %); load-based lag 58 %; tilt range L 40° vs R 41°; 45665 left samples in 330 strides; session N: left impact lags right by 49 % of the cycle (95 % CI 49–49 %); load-based lag 51 %; tilt range L 35° vs R 33°; 75997 left samples in 544 strides.

9 Sleep and the bedroom

The band recorded sleep from 04:35 to 10:03 (Fig. 22): 5.4 h asleep, the shortest night of the study, with 47 % light, 30 % deep and 23 % REM sleep and a mean heart rate of 64 bpm. The drinks at 01:00 and 02:32 fit with the participant being awake until late. In the bedroom (Fig. 23) $PM_{2.5}$ was about 100–110 $\mu g/m^3$ until 03:00 and then fell to about 40 $\mu g/m^3$ by 05:30, the same shape as on every monitored night but at about half the earlier level. Within the night heart rate correlated with $PM_{2.5}$ ($\rho = 0.38$) and negatively with sound ($\rho = -0.31$; Fig. 24); the $PM_{2.5}$ direction is opposite to day 6, which again indicates a shared time trend rather than an effect.

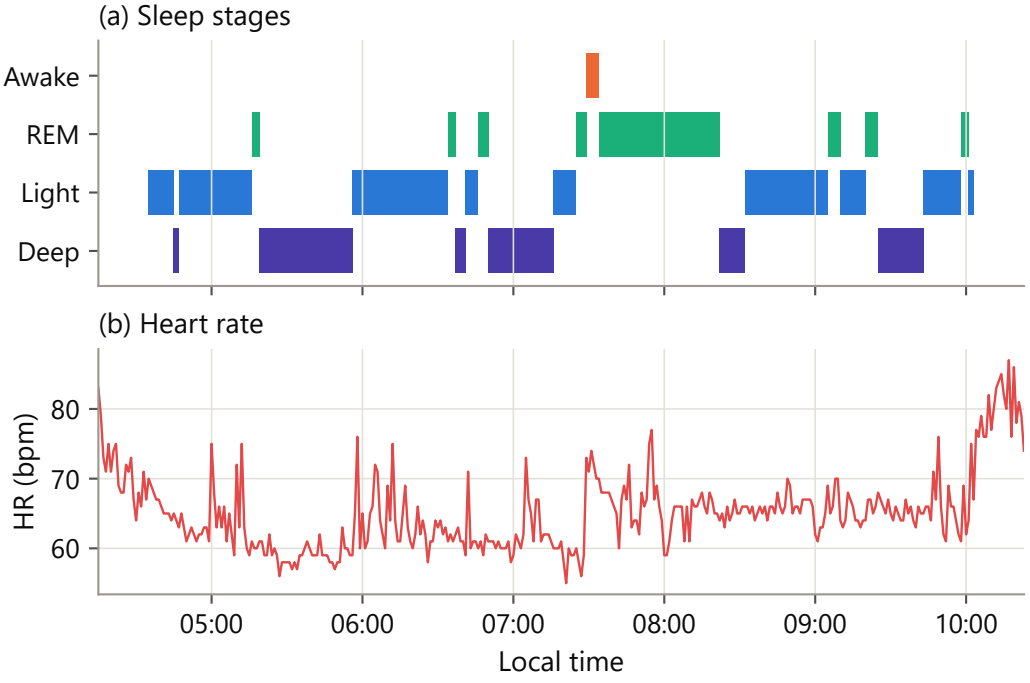


Figure 22: Sleep before day 7 from the wrist band. (a) Hypnogram, 04:35–10:03: 5.4 h asleep (efficiency 98 %), 47 % light, 30 % deep and 23 % REM. (b) Heart rate (mean 64 bpm, lowest 10-min mean 58 bpm). SpO_2 averaged 97.2 % (minimum 94 %).

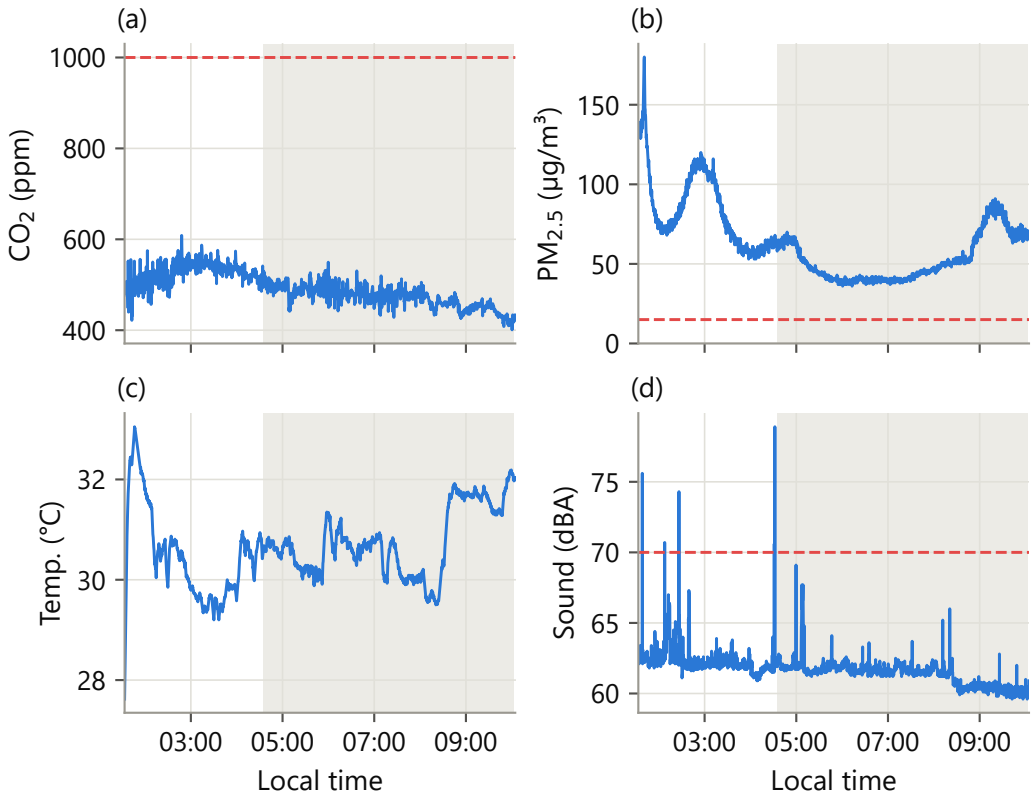


Figure 23: Bedroom environment overnight (session O, 01:33–10:04, 10 s averages): (a) CO_2, median 488 ppm (max 608); (b) $PM_{2.5}$, median 60 μg/m^3 (max 180); (c) temperature, median 30.6 °C; (d) sound level, median 62 dBA. Dashed lines: 1000 ppm CO_2, WHO $PM_{2.5}$ 24-h guideline, 70 dBA. The grey band marks sleep (04:35–10:03) from the wrist band. Temperature includes a -5 °C correction for sensor self-heating.

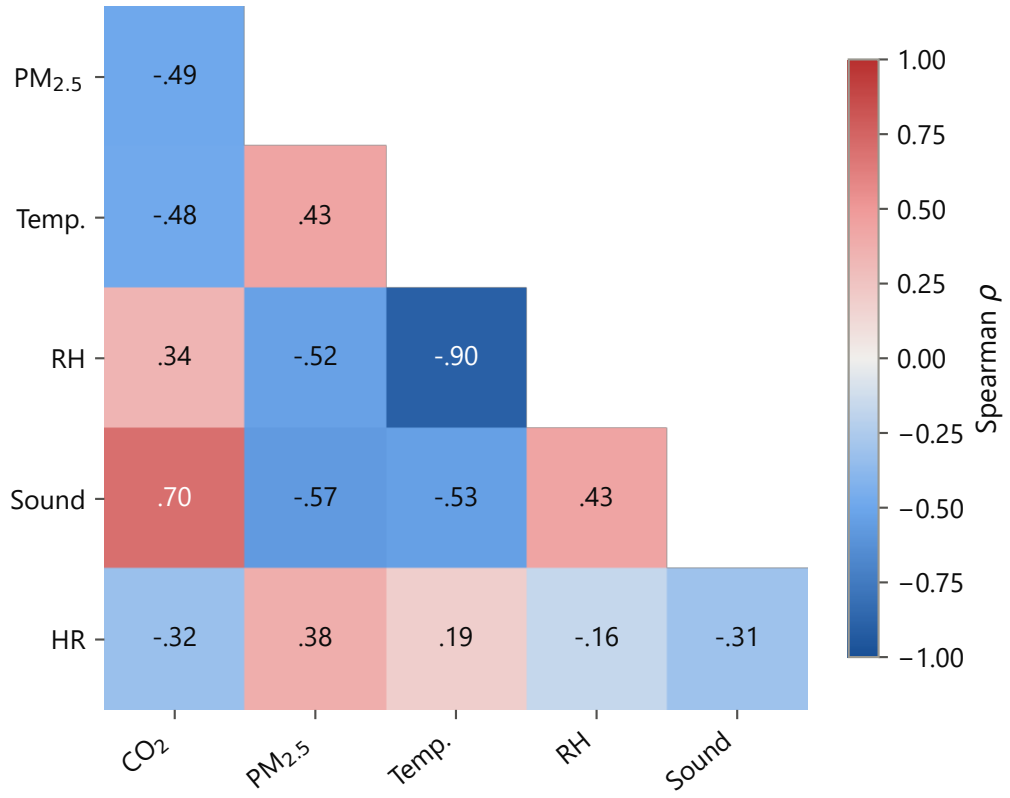


Figure 24: Correlations between bedroom environment and heart rate during sleep (session(s) O, sleep minutes only). Spearman rank correlations of 1-min means ($n = 329$–329 minutes per pair). Minute samples are autocorrelated, so coefficients are descriptive and no p-values are reported.

10 Lab and posture

The lab (Figs. 25–27) had the cleanest indoor air of the study: $PM_{2.5}$ 40 μg/m^3 (median), CO_2 490–770 ppm, 26.9 °C, 49 % relative humidity and a quiet 56 dBA. The posture cushion was connected for 22 minutes in total (Fig. 28): 82 % leaning backward and 18 % in the cushion's "straight" category.

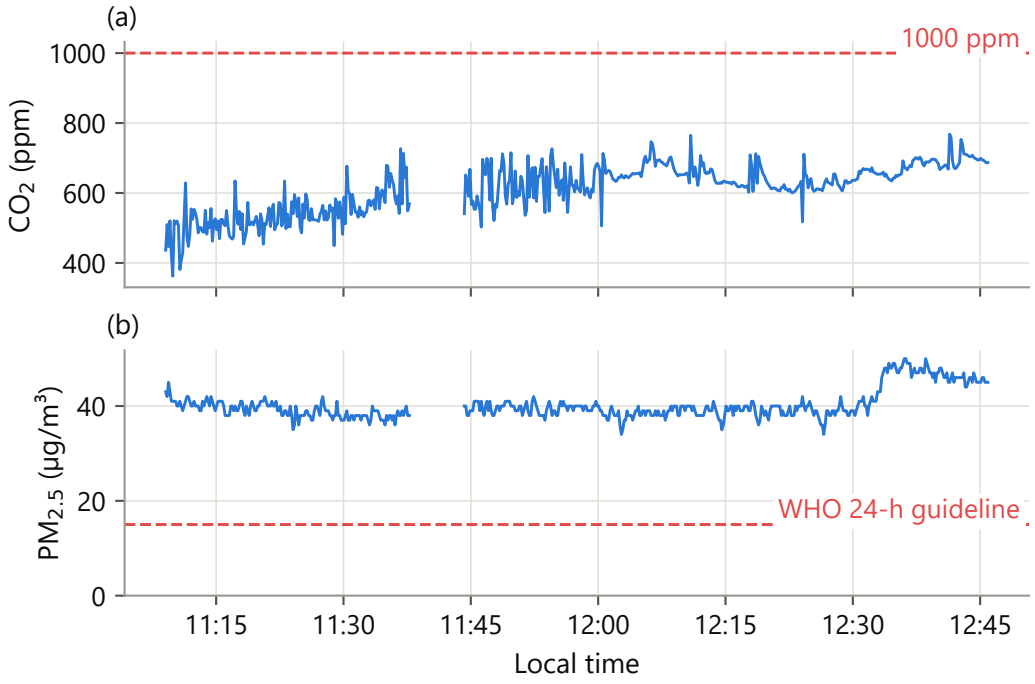


Figure 25: Indoor air quality from the environment node (10 s averages; session L, 11:09–12:45). (a) CO_2, median 628 ppm, above 1000 ppm for 0.0 % of the time. (b) $PM_{2.5}$, median 40 μg/m^3, above the WHO 24-h guideline (15 μg/m^3) for 100 % of the time. Line breaks mark periods when the node was disconnected.

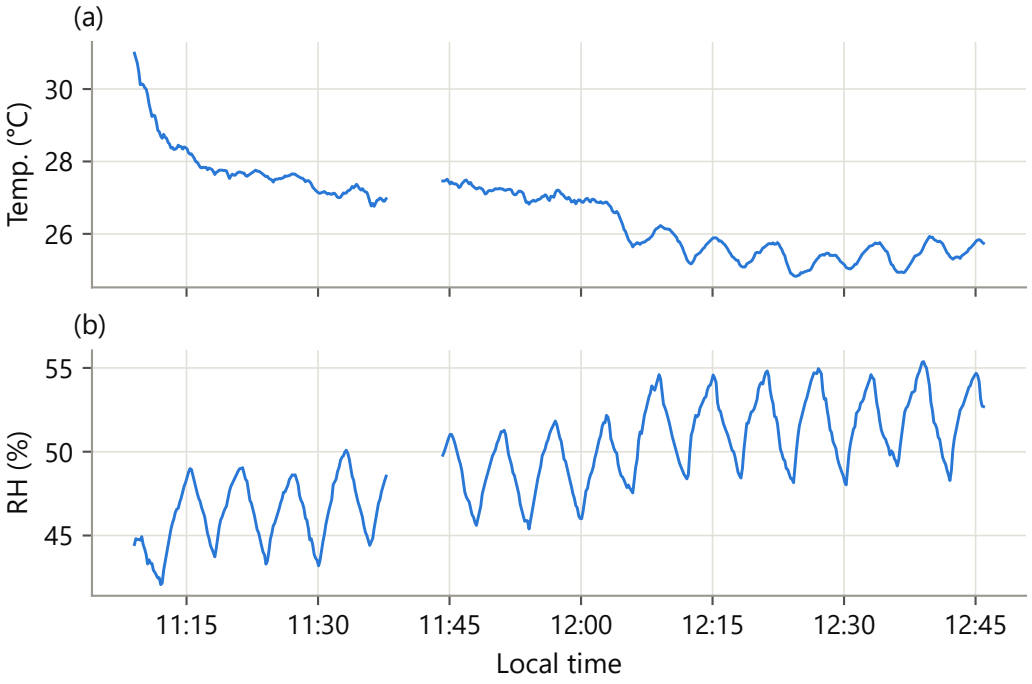


Figure 26: Indoor thermal conditions (session L, 11:09–12:45): (a) air temperature, median 26.9 °C (5–95th percentile 25.0–28.4 °C); (b) relative humidity, median 49 %. Temperature includes a -5 °C correction for sensor self-heating.

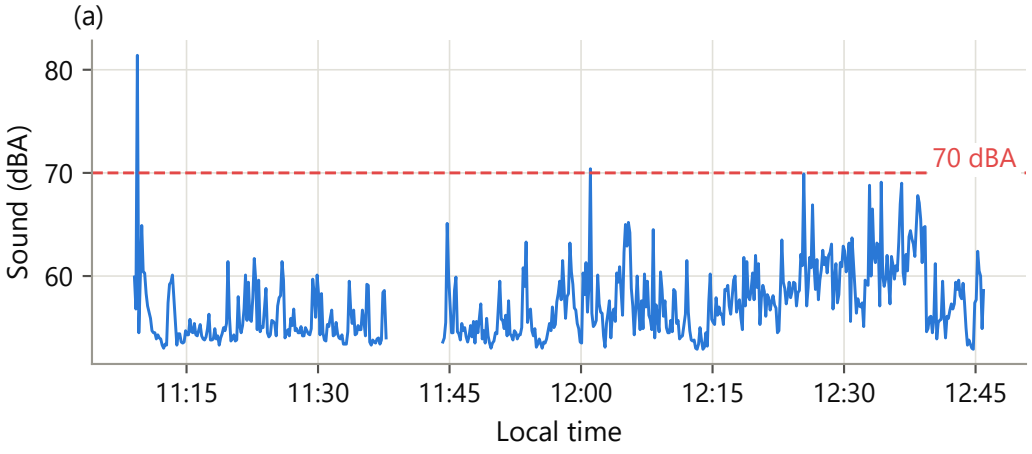


Figure 27: Indoor A-weighted sound level (session L, 11:09–12:45), median 56 dBA (95th percentile 63 dBA); above 70 dBA for 0 % of the time.

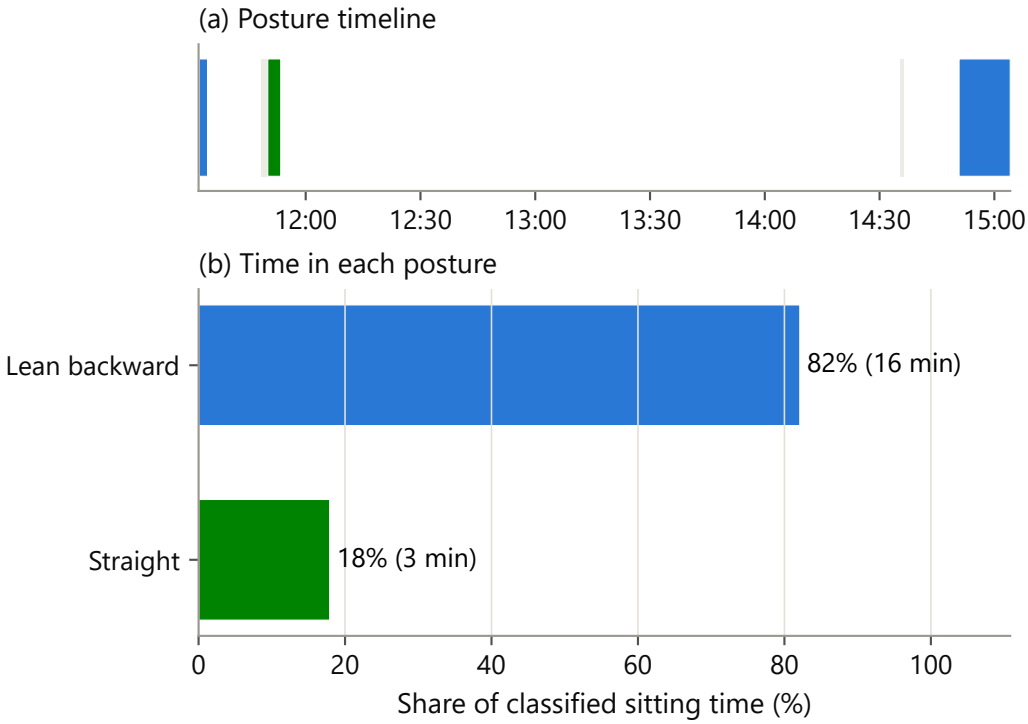


Figure 28: Sitting posture from the smart cushion. (a) Pose held over time; the cushion was connected during 11:32–11:34, 11:48–11:53, 14:35–14:36, 14:50–15:04 (22 of 471 min). Light grey = connected but no live pose event yet; white = disconnected. Onsets are shifted 10 s earlier to correct for the cushion's commit delay. (b) Share of the 19 classified minutes in each pose; median time between pose changes 3.4 min.

11 Comparison of days 1–7

Sleep (Fig. 29). Sleep lasted 6.1–7.5 h on days 1–6 and 5.4 h on day 7, which also had the latest onset (04:35). Heart rate during sleep was 60–64 bpm on every night.

Heart rate (Figs. 30 and 31). The night-time profiles are almost identical. Indoor sitting heart rate was highest on days 6 and 7 (about 104 bpm). Walking heart rate, excluding suspected cadence lock, ranged from 91 bpm (day 7) to 124 bpm (day 1).

Gait (Figs. 32 and 33). Over seven days gait was very stable: cadence 97–105 steps/min, stride time 1.14–1.24 s, speed 1.06–1.19 m/s, stride length 1.27–1.34 m, push-off timing 66–67 % and IMU stance 62–63 % of the stride on both feet (day 4 morning right: 60 %). On day 7 the IMU and pressure

Table 3: Headline metrics across days 1–7 (medians for environment values; left lag: left heel strike as % of the right gait cycle; †walking heart rate excludes minutes of suspected optical cadence lock, i.e. heart rate within 6 bpm of step rate for ≥2 consecutive minutes; stance and swing from IMU gait events; HR-lock min.: flagged / walking minutes).

Metric	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7
Time asleep (h)	6.6	6.3	6.6	6.1	7.5	7.4	5.4
Deep sleep (%)	27	36	35	28	28	32	30
REM sleep (%)	31	19	19	19	33	28	23
Mean HR asleep (bpm)	60	60	61	64	63	63	64
Mean awake HR (bpm)	94	88	92	98	96	104	96
HR > 120 bpm (min)	56	14	5	66	56	241	24
Steps (band)	7,173	7,044	5,661	7,021	4,863	8,641	4,284
Steps (phone)	8,109	6,608	5,422	8,492	5,131	8,435	4,694
HR–steps ρ	0.27	0.33	0.23	0.30	0.26	0.27	0.08
Walk cadence (steps/min)	105	101	102	100	97	102	100
Stride time (s)	1.14	1.19	1.18	1.19	1.24	1.17	1.20
Stride-time CV (%)	10.6	6.7	12.6	12.1	9.7	5.4	15.4
Stance R (%)	63	63	62	60	63	62	63
Stance L (%)	–	–	63	64	63	62	62
Swing R (s)	0.42	0.43	0.44	0.48	0.46	0.45	0.44
Swing L (s)	–	–	0.45	0.43	0.46	0.44	0.47
Push-off timing R (%)	66	66	66	62	66	66	66
Push-off timing L (%)	–	–	67	66	67	66	67
Upright sitting (%)	14	–	–	54	9	6	18
Walk distance (m)	925	1,824	1,449	1,290	917	624	747
Walk speed (m/s)	1.19	1.13	1.13	1.10	1.06	1.15	1.18
Stride length (m)	1.34	1.32	1.32	1.29	1.30	1.32	1.27
Walk mean HR† (bpm)	124	111	–	119	113	–	91
HR-lock min.†	0	0	22	5	3	7	3
Walking min.	16	35	30	17	20	11	10
Left lag (% cycle)	49.7	–	49.2	49.7	49.2	49.3	48.8
Indoor $PM_{2.5}$ ($\mu g/m^3$)	51	67	–	48	52	51	40
Indoor temperature (°C)	21.2	29.5	–	23.5	26.1	24.9	26.9
Indoor sound (dBA)	67	56	–	65	63	72	56
Outdoor $PM_{2.5}$ ($\mu g/m^3$)	59	71	101	68	68	71	57
Bedroom $PM_{2.5}$ ($\mu g/m^3$)	–	84	69	102	75	76	60
Fluid intake (mL)	1,744	1,292	2,158	1,660	890	1,967	1,265
Longest drink gap (h)	6.0	5.1	3.0	9.5	9.4	9.8	7.6

toe-offs on the right foot differed by 52–68 ms, as on days 5–6, confirming that the replacement Met1 sensor shifts the pressure-based toe-off while the IMU stance is unaffected.

Left versus right (Fig. 34). Step timing was 48.8–50.3 % in every session in which it could be measured.

Environment and hydration (Figs. 35 and 36). $PM_{2.5}$ exceeded the WHO 24-hour guideline in every setting on every day. Fluid intake from the smart-bottle log ranged from 0.9 to 2.2 L per day (day 7: 1.3 L), and the longest gap without a drink was 9.4–9.8 h on days 4–6 and 7.6 h on day 7.

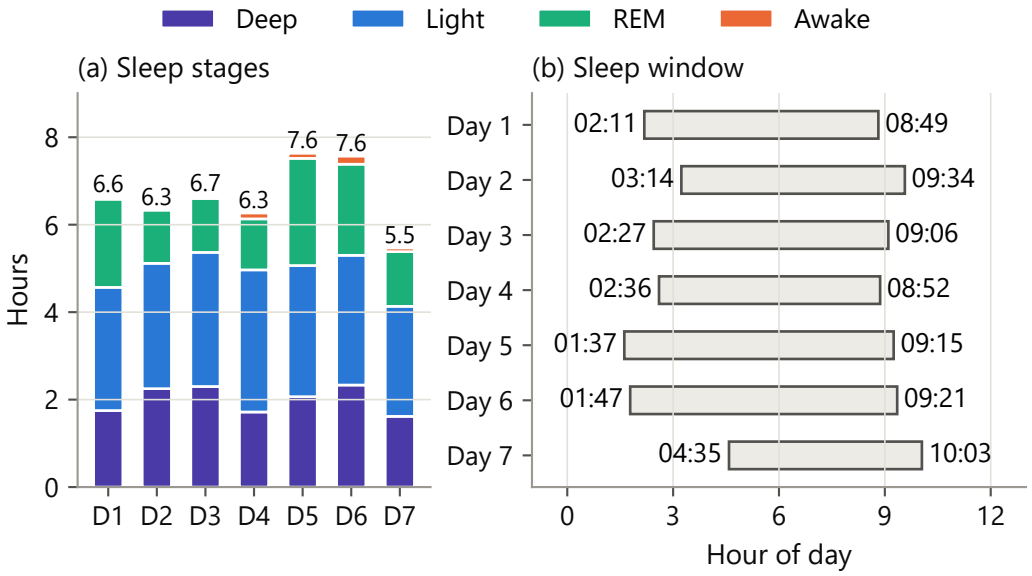


Figure 29: Sleep on the night before each day (wrist band): (a) time in each stage and total; (b) sleep onset and wake time.

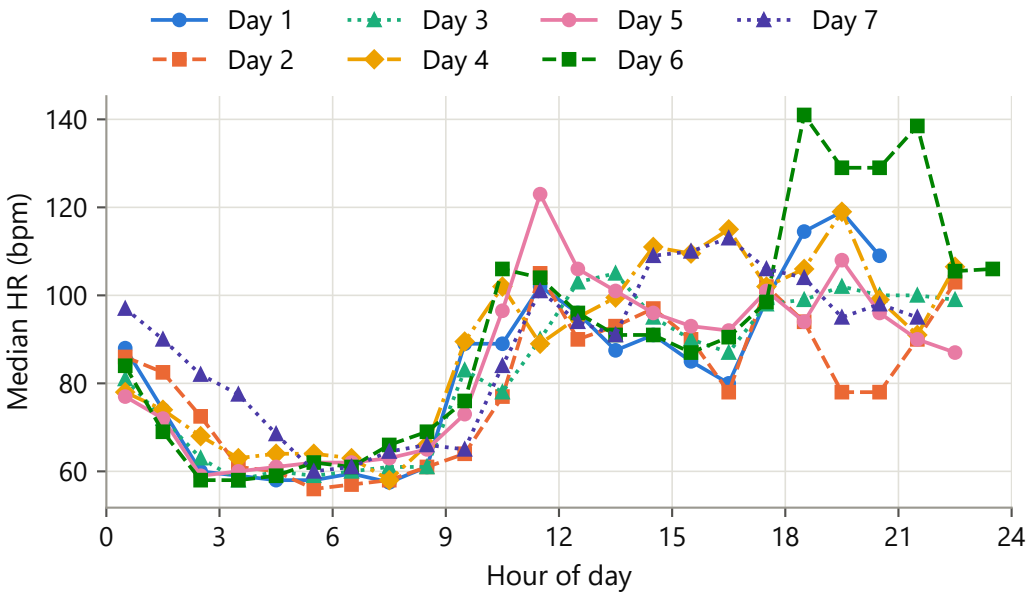


Figure 30: Hour-by-hour median heart rate (wrist band) on each day. Hours without band data are not drawn.

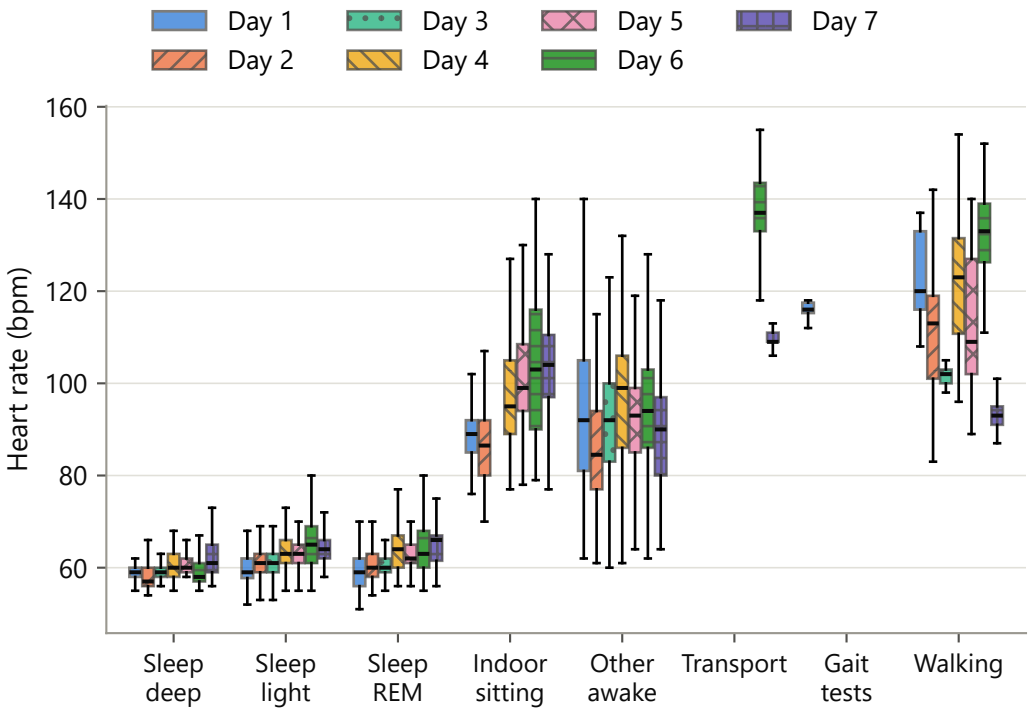


Figure 31: Heart rate by context on each day (boxes: inter-quartile range and median; whiskers 1.5 IQR). Contexts are defined in the same way every day: sleep stages from the wrist band, sensor sessions from the session log, and all other awake minutes.

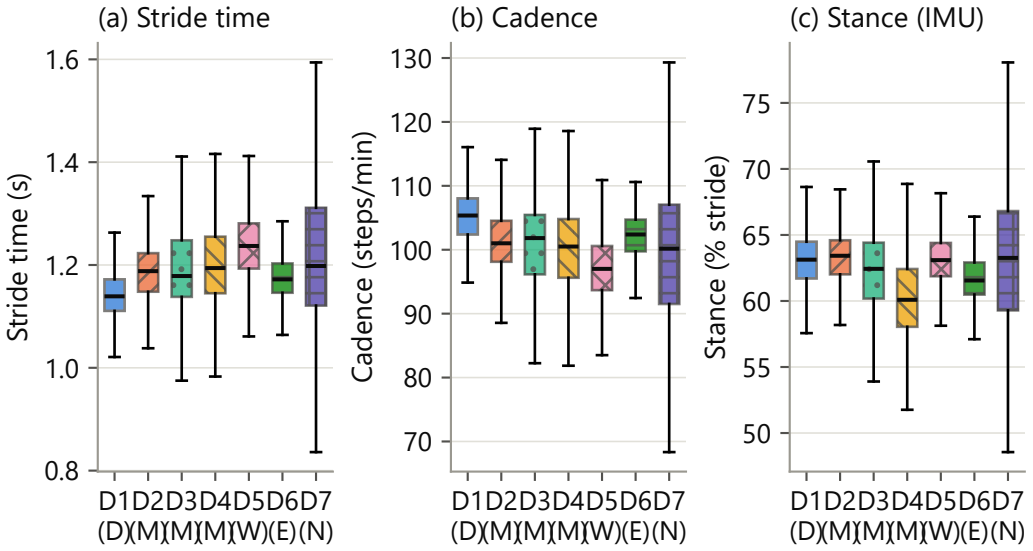


Figure 32: Right-foot gait during the main walk of each day (steady walking only: bouts of at least 10 strides without their first and last two strides; day 1: session D, 896 steady strides; day 2: session M, 1842 steady strides; day 3: session M, 1560 steady strides; day 4: session M, 912 steady strides; day 5: session W, 983 steady strides; day 6: session E, 599 steady strides; day 7: session N, 656 steady strides). Cadence is 120 divided by stride time. Stance is measured from IMU gait events (heel-strike impact to IMU toe-off), which stay valid after the right first-metatarsal pressure sensor was replaced (day 5 onwards).

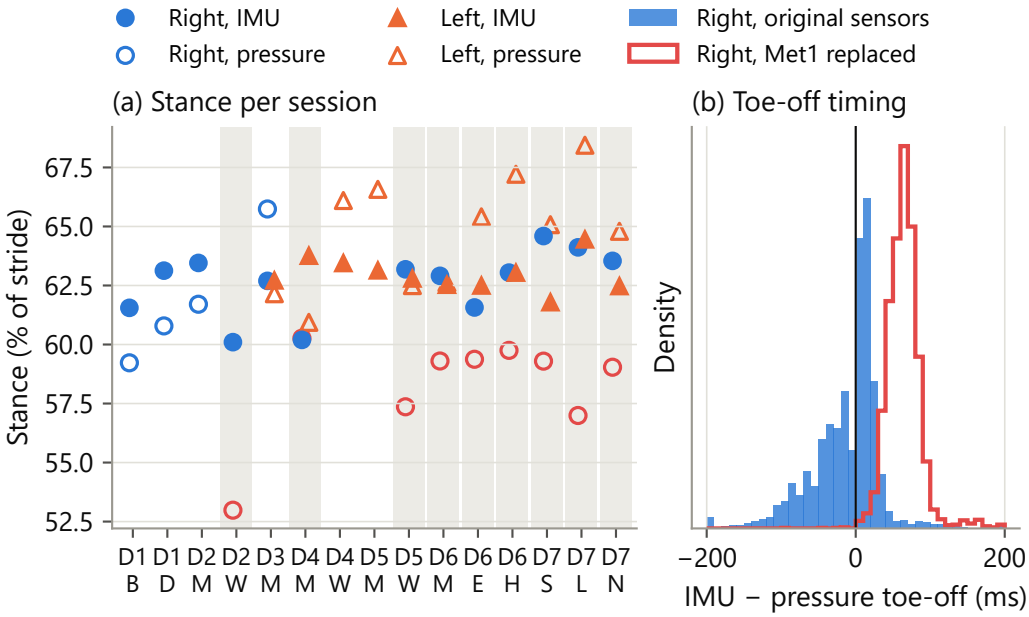


Figure 33: IMU-based toe-off and stance. (a) Median stance of steady strides in every gait session with at least 100 steady strides, from IMU gait events (filled: heel-strike impact to the instant the push-off rotation rate of the foot has fallen to half its peak) and from the pressure sensors (open; red where the right first-metatarsal sensor was faulty or replaced, grey background). (b) Per-stride difference between the IMU and pressure toe-off on the right foot with original, healthy sensors (D1-B, D1-D, D2-M, D3-M; 4241 strides): median +0 ms, SD 46 ms, inter-quartile range -34 to 16 ms. After the right first-metatarsal sensor was replaced (red outline, 3117 strides) the pressure toe-off came 63 ms earlier than the IMU toe-off, so the pressure-based right stance was too short (red open circles in a), while the IMU stance was unchanged.

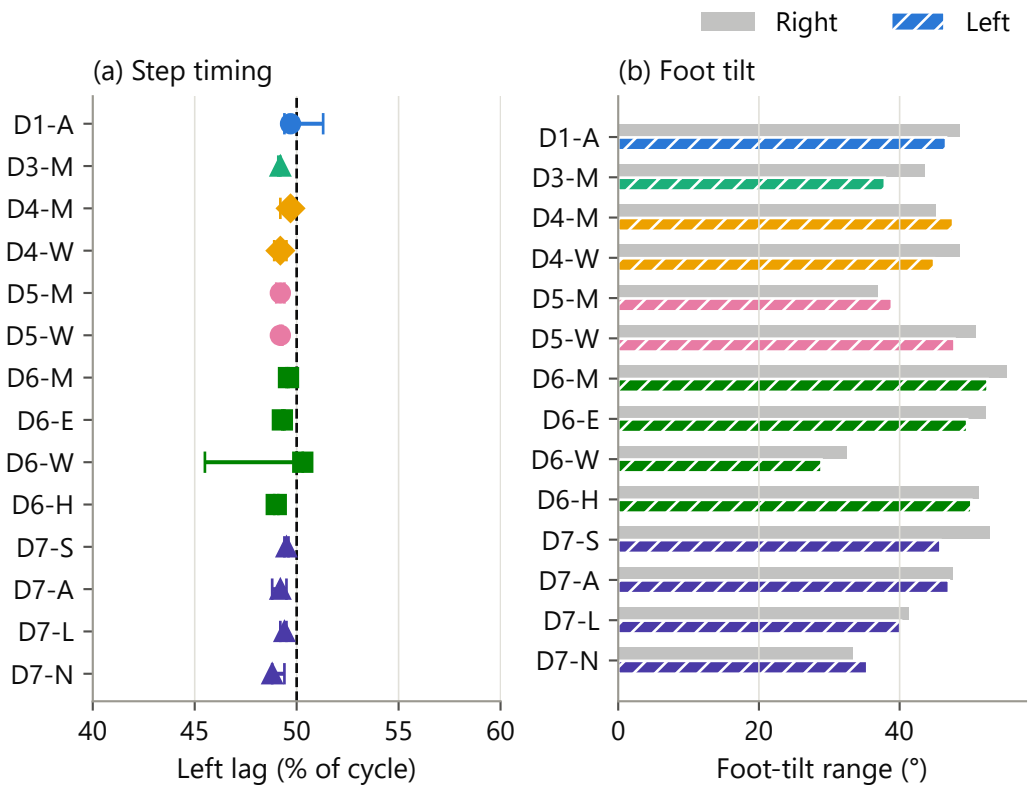


Figure 34: Left/right comparison in every session where the left insole delivered enough data (labels: day and session). (a) Lag of the left heel-strike impact behind the right one, as a percentage of the gait cycle, with bootstrap 95 % confidence intervals; 50 % (dashed) is perfectly symmetric step timing. (b) Range of foot tilt over the cycle for each foot. Sessions where the left and right acceleration curves did not match in shape are omitted.

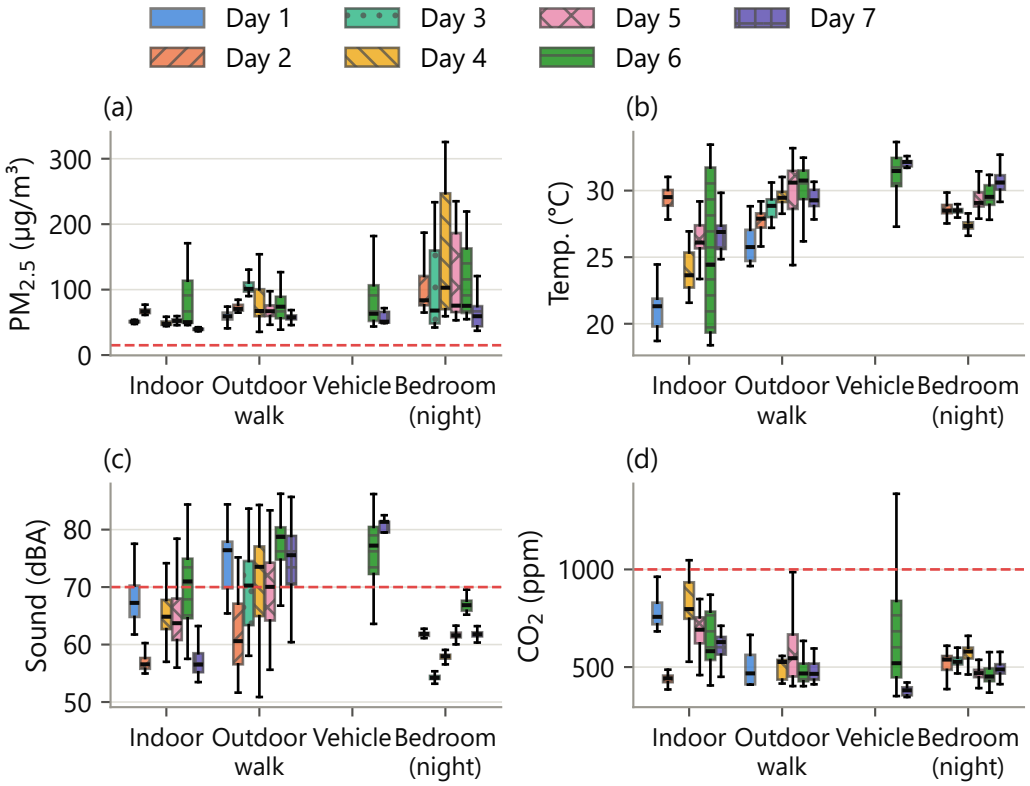


Figure 35: Environment-node readings (1-min means) by setting and day: (a) $PM_{2.5}$, (b) temperature, (c) sound level and (d) CO_2; outdoor CO_2 readings below 400 ppm are excluded as implausible. Dashed lines: WHO $PM_{2.5}$ 24-h guideline, 70 dBA, 1000 ppm. A setting is missing on days without that recording. Temperatures include a -5 °C self-heating correction.

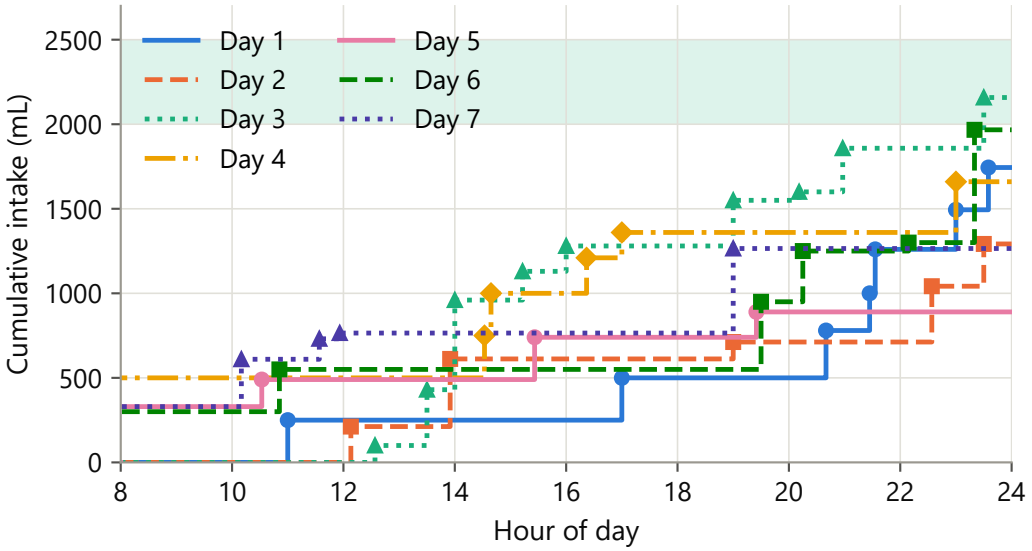


Figure 36: Cumulative fluid intake from the smart-bottle log on each day (plus drinks detected in the bottle's data stream on day 7). The shaded band is the EFSA adequate intake for total water (2.0–2.5 L/day including food).

12 Water before walking and heart rate

On day 7 the participant drank before both walks to test whether drinking beforehand changes heart rate while walking. All walks of the study were used (Fig. 37; `water_walk.csv`). For each walk the median heart rate of its walking minutes was taken, excluding suspected cadence-lock minutes. The day-3 morning walk and the day-6 walk to the metro were left out because most of their walking minutes were locked, and the morning walks of day 7 were too short (under 4 walking minutes). That leaves 10 walks. Three of them followed at least 200 mL in the previous 2 hours (day 6 W and H, day 7 N).

- *Walking heart rate* was not related to the water drunk in the 2 hours before ($\rho = 0.27$, 95 % CI -0.59 to 0.83), to the time since the last drink ($\rho = -0.22$) or to the water drunk so far that day ($\rho = 0.09$). Allowing for temperature, each 100 mL drunk in the 2 hours before changed walking heart rate by -0.1 bpm (CI -6.2 to $+8.8$ bpm), i.e. no detectable effect. The two late walks on day 6 had high heart rates (135 and 147 bpm) despite 300–700 mL beforehand, and the day-7 evening walk had the lowest (92 bpm) after 500 mL.
- *Rise of heart rate above rest* (walking heart rate minus the resting value 5–30 minutes before) was smaller when more water had been drunk beforehand ($\rho = -0.61$, CI -0.90 to 0.18; water so far that day $\rho = -0.76$, CI -1.00 to -0.16). The day-7 evening walk was the only walk with heart rate *below* the resting value before it (-8.5 bpm). However, the rise is also strongly related to the time of day ($\rho = -0.71$): evening walks start from a higher resting heart rate and naturally come after more drinking. Allowing for time of day, the association weakens to a partial ρ of -0.34 (2-hour water) and -0.48 (water so far that day).

Conclusion. Within this study, drinking before a walk did not measurably change walking heart rate. A smaller rise above rest after drinking is consistent with the participant's hypothesis, but it is largely explained by the time of day, and ten walks are too few to separate the two. A direct test would alternate "drink 500 mL" and "no drink" before the same route at the same time of day over several days.

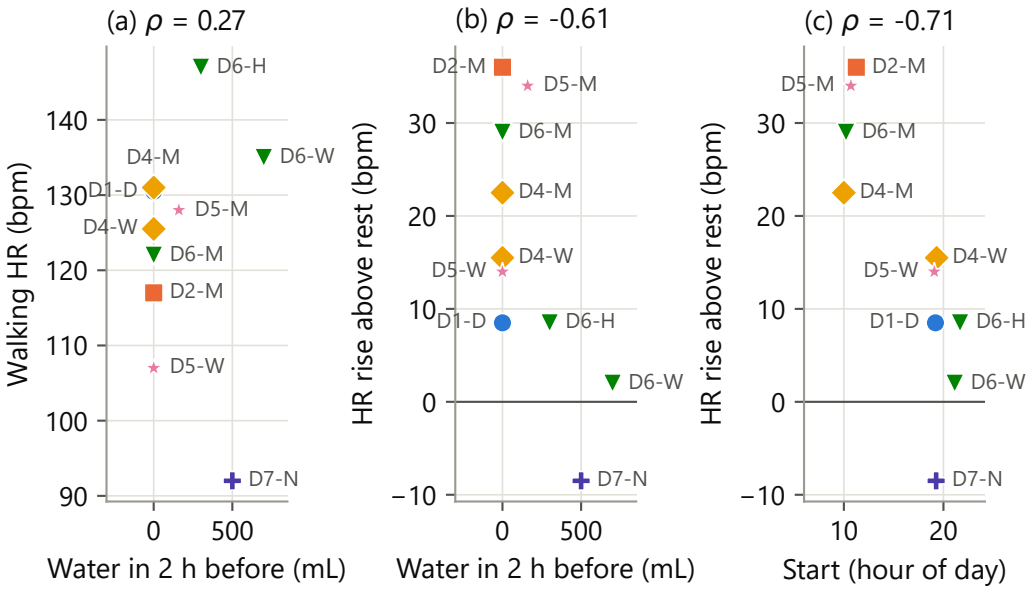


Figure 37: Water drunk before walking and heart rate during the walk (10 walks over days 1–7; median heart rate of walking minutes excluding suspected cadence lock). Walks D3-M, D6-E are excluded because most of their walking minutes were suspected cadence lock. (a) Walking heart rate against fluid from the bottle log in the 2 hours before the walk; (b) rise of heart rate above the resting value 5–30 minutes before the walk, against the same; (c) that rise against the time of day. Spearman ρ with bootstrap 95 % CI: (a) 0.27 (-0.59 to 0.83), (b) -0.61 (-0.90 to 0.18), (c) -0.71 (-0.89 to -0.17); (b) after allowing for the time of day: partial $\rho = -0.34$. Labels: day and walk.

13 Correlation and covariance across seven days

How robust is the relationship between heart rate and step rate? "Robust" here means *reproducible*, not strong. It was checked in five ways (Fig. 39; details in the file `hr_steps_robustness.json`):

1. *Every day:* the correlation was positive on all 7 days ($\rho = 0.08$–0.33). If there were no relationship, 7 positive days out of 7 would occur by chance with probability 0.016 (two-sided sign test). The 95 % block-bootstrap interval excluded zero on 6 of the 7 days; day 7 was borderline (0.00 to 0.17).

2. *No single day drives it:* leaving out each day in turn gives a pooled

Table 4: Relationships between per-minute measurements, days 1–7. Pairs are listed as heart rate (or the first variable named) against the second. Per day: Spearman ρ on each day. Level: pooled within-day (repeated-measures) Pearson r with 10-minute block-bootstrap 95 % CI. Change: the same for minute-to-minute changes (*: CI excludes zero). Consistent: same sign with $|\rho| > 0.2$ on every day, both CIs excluding zero and $|r| \geq 0.3$; level only: consistent in levels but not in changes; varies: pooled CI excludes zero but per-day signs or sizes differ. Covariances and regression slopes are given in relationships.csv.

	Per-day ρ									
Pair	D1	D2	D3	D4	D5	D6	D7	Level r (95 % CI)	Change r	Verdict
Heart rate, awake										
Steps	.27	.33	.23	.30	.26	.27	.08	.29 (.24, .34)	−.04*	varies
Heart rate, walking										
Cadence	.16	.00	–	.19	.25	−.08	.07	.09 (−.10, .25)	−.02	none
Speed	−.10	−.18	–	−.02	.34	.20	.05	.19 (−.13, .43)	−.05	none
Temperature	−.78	.59	–	.67	.42	−.52	.34	.32 (−.02, .62)	.12	none
$PM_{2.5}$	−.12	−.20	–	−.25	−.33	.37	−.45	−.21 (−.35, .00)	−.13	none
Sound	.01	.07	–	−.11	.28	.16	−.46	.05 (−.22, .31)	.09	none
Heart rate, asleep										
Temperature	–	.38	.00	.11	.16	.51	.19	.24 (.16, .33)	−.03	varies
CO_2	–	−.32	.19	.04	−.11	.02	−.32	−.09 (−.17, .00)	−.01	none
$PM_{2.5}$	–	−.23	.12	.38	−.36	−.59	.38	−.04 (−.12, .04)	−.01	none
Sound	–	−.32	−.10	.42	−.33	−.45	−.31	−.23 (−.32, −.14)	.01	varies
Gait and sensors, walking										
Stride vs cadence	−.29	−.55	−.89	−.68	−.50	−.69	−.38	−.41 (−.59, −.25)	−.32*	consistent
Insole vs band	.42	.74	.51	.60	.42	.60	.16	.35 (.20, .49)	.03	varies
Environment										
Sound vs $PM_{2.5}$	.23	.60	.72	.31	−.17	.27	.23	.31 (.18, .42)	−.01	varies
PM_{10} vs $PM_{2.5}$	.96	1.00	1.00	1.00	1.00	1.00	1.00	1.00 (1.00, 1.00)	.99*	consistent
RH vs temperature	.09	−.93	−.90	−.08	.08	−.64	−.24	.22 (.11, .32)	−.13*	varies

within-day r of 0.27–0.31.

3. *Not an artefact of cadence lock:* minutes in which the band may have reported the step rate instead of the pulse would inflate the correlation. Removing the 42 suspected minutes gives $r = 0.30$ (CI 0.24–0.35).

4. *Autocorrelation:* the confidence interval resamples 10-minute blocks within each day, so neighbouring, similar minutes are not counted as independent evidence. The interval is still 0.23–0.34.

5. *Time scale:* minute-to-minute changes are unrelated ($r = -0.03$), because heart rate responds to activity over several minutes. Changes of 10-minute means correlate with $r = 0.23$ and of 30-minute means with $r = 0.42$.

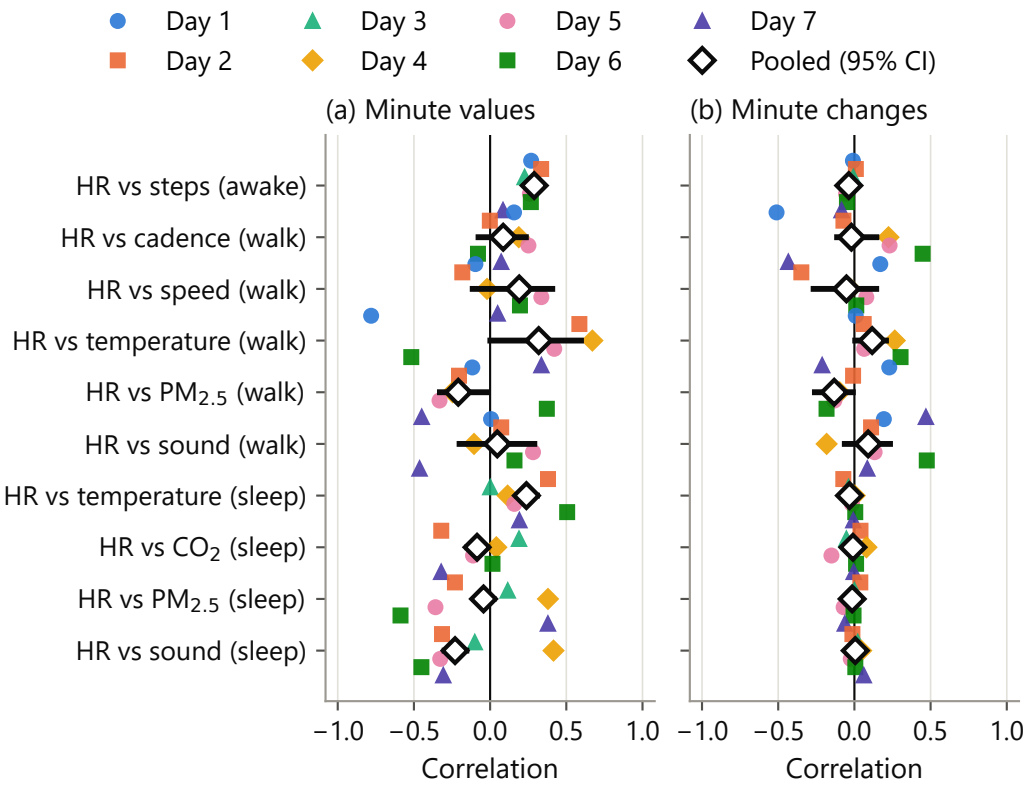


Figure 38: Heart-rate relationships on each day (coloured symbols: per-day Spearman ρ) and pooled over days (diamond and bar: within-day Pearson r with 95 % block-bootstrap CI). (a) Minute values; (b) minute-to-minute changes, which remove slow trends that two variables share only because both drift over a session. Walking minutes are those with at least 60 insole steps, excluding minutes of suspected optical heart-rate cadence lock; sleep minutes are those inside the band's sleep period during the overnight bedroom recording.

The relationship is weak, however. Step rate explains about 8 % of the variation in heart rate within a day (about 20 bpm per 100 steps/min), because most awake minutes are sitting, when heart rate varies for other reasons (heat, hydration, posture, stress). It is the only heart-rate relationship in the study that passes all five checks.

- *Heart rate and step rate*: see above (pooled $r = 0.29$).
- *Walking heart rate and $PM_{2.5}$* was negative on five of six days (pooled $r = -0.21$, CI -0.35 to 0.00). As discussed before, this is most likely

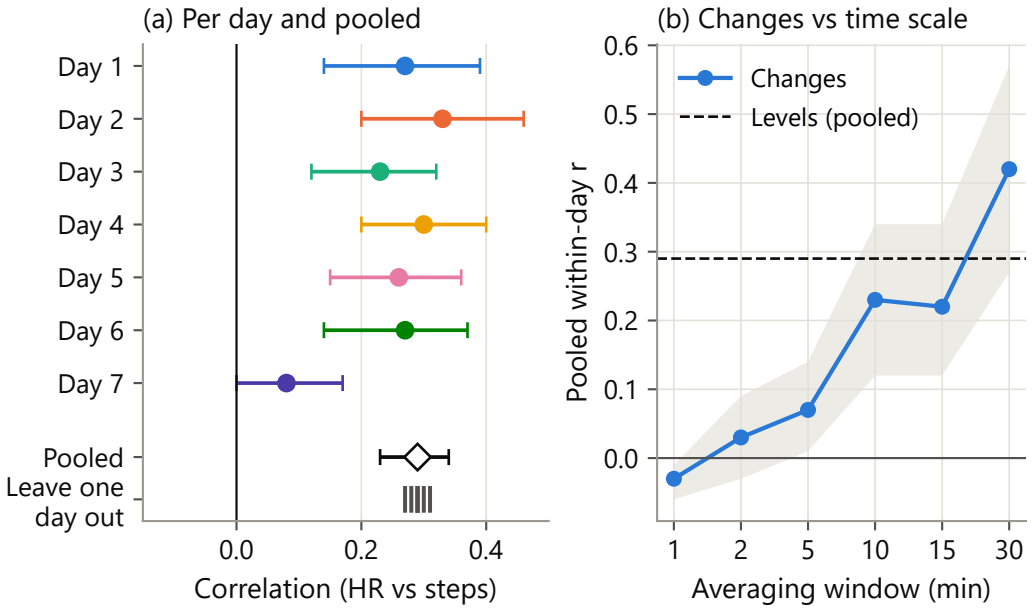


Figure 39: Robustness of the relationship between wrist heart rate and step rate in awake minutes (6,368 minutes, days 1–7). (a) Spearman ρ per day with 10-minute block-bootstrap 95 % CIs; pooled within-day Pearson r (diamond) and the pooled r when each day in turn is left out (ticks: 0.27–0.31). (b) Pooled within-day correlation between *changes* of heart rate and step rate after averaging over windows of 1 to 30 minutes (band: 95 % CI); dashed: correlation of levels. Minute-to-minute changes are unrelated ($r = -0.03$) because heart rate responds to activity over several minutes; over 30-minute windows the changes correlate with $r = 0.42$.

confounded by pace and location.

- *Walking heart rate and temperature* changed sign between days (from -0.78 to $+0.67$), so it is not consistent.
- *Heart rate during sleep* was positively related to bedroom temperature (pooled $r = 0.24$, CI 0.16–0.33) and negatively to sound ($r = -0.23$, CI -0.32 to -0.14), but neither holds for minute changes. Both are most likely shared night-time trends (the room cools and quietens while heart rate falls in deep sleep).
- *Checks:* stride time fell with cadence and $PM_{2.5}$ and PM_{10} were practically identical on every day.

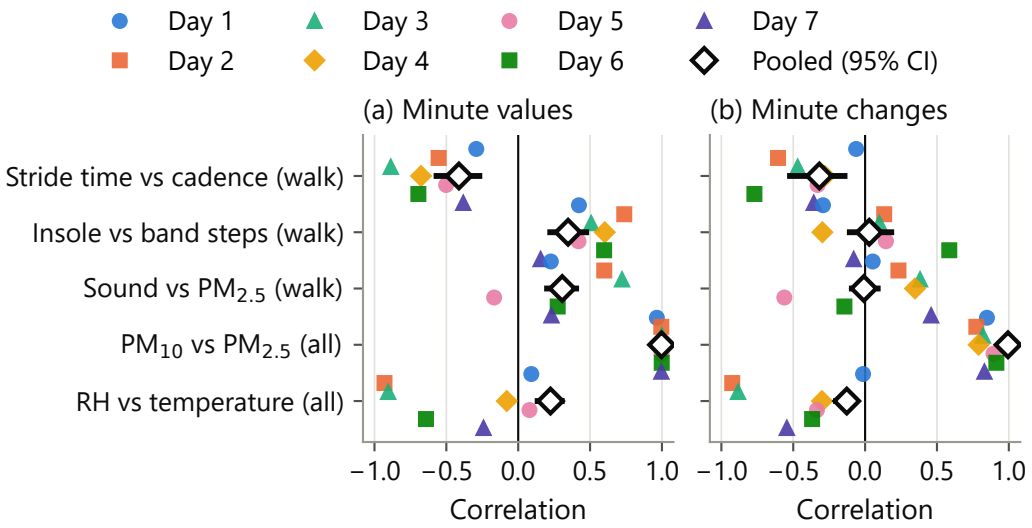


Figure 40: Gait, sensor-agreement and environment relationships on each day and pooled, presented as in Fig. 38. These pairs act partly as checks: stride time must fall with cadence, and $PM_{2.5}$ and PM_{10} come from the same particle sensor.

14 Data quality and next steps

1. **Smart bottle**: the fused channel is accurate to about 2 mL and detected both confirmed drinks; the capacitive channel reads about 18 % low and its 500 mL value must be discarded. Keeping the bottle connected all day would replace the read-off log with a continuous record.

2. **Right insole**: re-solder the heel sensor; Met1 is still the replacement type.

3. **Soldering**: use fume extraction or keep a distance from the iron; the node measured more than 3,333 $\mu g/m^3$ within seconds.

4. **Next**: the overall 7-day report and the machine-learning stage.